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Discoveries

Clean Breaks: Physical Force Field Deconvolution of Recombination Across Sequence Manifolds

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Abstract

Homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages. Reticulate evolution shatters bifurcating trees. Existing detection methods remain trapped between computationally prohibitive phylogenetic changepoint algorithms that scale exponentially with breakpoint count and isolated triplet heuristics that trigger high false positive rates under mutational rate variation. We introduce `RhizAeon`, an open-source framework that models reticulation as continuous trajectory mechanics in empirical metric spaces without evaluating trees or enumerating triplets. By coupling continuous phylogenetic cross-attention over landmark lineages with an unpolarized ancestral root token that absorbs autapomorphic noise, `RhizAeon` deconvolves reticulation through physical force field dynamics in linear time. Across extensive coalescent benchmarks, operational stress tests, and large-scale parameter sweeps, `RhizAeon` maintains strict false positive control on non-recombinant nulls while sustaining throughput exceeding 30 genomes per second. Empirical applications across wildlife Sarbecovirus spike chimeras, bacterial chromosome mosaicism in Streptococcus pneumoniae, and primate HLA Class I immunogenomics resolve deep reticulations and fine-scale micro-conversions in fractions of a second. The platform executes client-side surveillance within standard web browsers without server data transfer.

Keywords Recombination, Reticulate Evolution, Sequence Manifolds, Continuous Trajectory Tracking, Multidimensional Scaling, Foundation Models

Introduction

Homologous recombination and horizontal gene transfer assemble mosaic chromosomes from divergent evolutionary lineages (Simon-Loriere and Holmes, 2011; Didelot and Maiden, 2010). By splicing divergent genetic material into a single chromosome during co-infection, genetic exchange circumvents the slow waiting times imposed by clonal interference. Point mutations accumulate gradually. Recombination generates instantaneous evolutionary leaps, combining distant adaptive alleles and breaking genome-wide linkage disequilibrium within a single replication cycle. In viral and bacterial epidemics, mosaicism alters transmission dynamics, shifts antigenic presentation, and accelerates the acquisition of antimicrobial resistance determinants (Alfonsi et al., 2024; Didelot and Maiden, 2010). These dynamics govern circulating recombinant forms of human immunodeficiency virus (HIV-1), recombinant variants of SARS-CoV-2, and multi-drug resistant bacterial clones. In eukaryotes, meiotic crossover and adaptive introgression purge deleterious mutational load, drive ecological speciation, and assemble complex hybrid phenotypes across plant and animal taxa (Rieseberg et al., 2003; McVean et al., 2004). Resolving physical crossover boundaries remains foundational across evolutionary biology.

Molecular phylogenetics presumes that sequence evolution follows bifurcating trees. Reticulate evolution shatters trees. Because distinct genomic segments inherit conflicting ancestral genealogies, forcing mosaic alignments into a single bifurcating phylogeny distorts every downstream evolutionary inference (Schierup and Hein, 2000; Kosakovsky Pond et al., 2006). Topological reconstructions manufacture artificial clades, compress basal divergence times, and spuriously inflate terminal branch lengths. Downstream molecular clocks fail just as severely. Reticulate histories violate substitution rate constancy across sites, corrupting evolutionary rate estimates, node calibrations, transmission cluster reconstructions, and phylodynamic projections. In comparative codon models, reciprocal exchanges generate apparent homoplasies that map to multiple tree branches, mimicking convergent amino acid substitutions. Standard selection tests, including MEME and BUSTED (Murrell et al., 2012, 2015), mistake reticulate homoplasies for diversifying positive selection, inflating false positive rates across neutral codons. Partitioning mosaic alignments into tree-congruent segments safeguards all downstream evolutionary inference.

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Existing detection tools remain split between two incompatible paradigms (Posada, 2002; Martin et al., 2015). Phylogenetic changepoint methods, exemplified by GARD (Kosakovsky Pond et al., 2006), formulate crossover detection as model selection across partitioned trees under continuous-time Markov substitution models. These methods optimize partition boundaries through likelihood ratio criteria while fitting independent branch lengths across candidate segments. The computational arithmetic is punishing. Because the joint combinatorial space of partition boundaries and tree topologies scales exponentially with breakpoint count and cubically with taxon count ($\mathcal{O}(L^K \cdot N^3)$ for K breakpoints across N taxa of length L), multi-breakpoint searches across hundreds of genomes require days of cluster compute. Sequence heuristics take the opposite extreme. Suites such as 3Seq (Lam et al., 2018) and RDP5 (Martin et al., 2021) scan sequence triplets for localized clustering of polymorphic mutations within sliding windows. They evaluate rapidly, yet they operate without a global phylogenetic reference frame. When substitution rates vary across sites and lineages, localized mutational clustering mimics reciprocal genetic exchange. Under realistic Gamma rate heterogeneity, this mutational clustering inflates heuristic false positive rates beyond thirty percent (Posada and Crandall, 2001; Kosakovsky Pond et al., 2006). When recombinant segments descend from unsampled ancestral lineages, heuristics either fail entirely or misidentify recombinant lineages as parental donors, placing arbitrary point boundaries inside flat, uninformative genomic plateaus without confidence intervals.

Resolving reticulate evolution at genome scale requires abandoning discrete trees while retaining topological protection against mutational rate variation. Genomes trace continuous trajectories through sequence space. Scanning from 5' to 3' along an alignment, clonal lineages maintain stable relative divergence, flowing through sequence space like parallel currents. A recombinant chromosome breaks this parallel flow. It travels within the current of one parental lineage, then deflects sharply across sequence space into an alternative parental current at the crossover junction. We introduce `RhizAeon`, an open-source framework that models reticulate evolution through continuous trajectory mechanics and zero-drift force field dynamics in linear time ($\mathcal{O}(NKL)$ for K reference landmarks). Its name derives from the Greek *rhiza* (root), evoking the reticulated, interconnected network of the biological rhizome across deep evolutionary time (*aeon*) to supplant strictly bifurcating arboreal trees. By measuring instantaneous sequence affinity against reference lineages and absorbing lineage-specific rate variation into an unpolarized ancestral root sink, `RhizAeon` transforms recombination detection into physical trajectory deconvolution. Clonal evolution produces balanced thermal fluctuations that cancel out as a centered Brownian bridge. Recombination breaks this physical equilibrium, imparting sustained directional momentum that pinpoints crossover boundaries and consolidates co-recombinant transmission swarms into unified ancestral events.

We validated `RhizAeon` across coalescent benchmarks, specialized operational stress tests, and a grand continuous parameter sweep of 150,000 whole genomes. Across the canonical 3,200-trial coalescent benchmark of Posada and Crandall (2001), `RhizAeon` maintained an empirical false positive rate of 0.58% under extreme Gamma rate heterogeneity while recovering calibrated ancestral event counts. In operational stress tests modeling epidemic swarms, multi-parent cascades, ghost donor introgression, and micro-conversions down to 50 base pairs, trajectory deconvolution eliminated the combinatorial false alarms that cripple pairwise heuristics. Empirical evaluations across 30.9 kb Sarbecovirus genomes, the 2.22 Mb multidrug-resistant *Streptococcus pneumoniae* PMEN1 chromosome, and 379 primate *HLA* Class I alleles demonstrate that `RhizAeon` resolves complex whole-genome mosaicism and polarized Ancestral Recombination Graphs in fractions of a second without taxon subsampling.

The RhizAeon Framework: Principles of Operation

`RhizAeon` detects reticulate evolution by tracking continuous lineage trajectories through sequence space. Scanning along an alignment from 5' to 3', a non-recombinant genome maintains a stable genetic identity: its closest relatives remain its closest relatives, tracing parallel paths through sequence space. Recombination breaks this stability. A mosaic genome travels in the company of one parental lineage, then abruptly shifts allegiance to an alternative donor across the exchanged segment. To capture these shifts in linear time without constructing trees or sliding windows, `RhizAeon` measures the instantaneous affinity of each query sequence against a representative panel of reference lineages at every polymorphic site.

To evaluate affinity at each polymorphic site, `RhizAeon` adapts cross-attention, the foundational mechanism of transformer neural networks. The query sequence acts as the query, while the reference lineages act as keys. At every position along the genome, the query sequence evaluates which reference lineage it resembles, asking essentially: which lineage should I attend to right now? Rather than treating all nucleotide matches equally, attention scores are weighted by continuous-time Markov substitution probabilities: sharing a rare derived allele commands strong attention, while common transitions command less. To prevent private mutations from creating spurious matches, the reference panel includes an ancestral root token that absorbs alleles absent from every contemporary reference. A softmax function then normalizes these scores into an attention distribution that sums to one across all reference lineages at each position. Because individual sites fluctuate stochastically, `RhizAeon` does not evaluate positions in isolation or smooth them with sliding windows. Instead, it subtracts each lineage's genome-wide background attention and tracks the cumulative deviation along the chromosome. In non-recombinant regions, positive and negative fluctuations cancel out, wandering near zero like a random walk. When a sequence enters a recombinant tract, sustained attention to the donor drives the cumulative curve steadily upward. At the crossover junction, donor attention ceases and home attention resumes, causing the cumulative curve to peak and descend. A single crossover forms a sharp triangular arch, whereas an inserted cassette produces a trapezoidal plateau. The switch points are located directly at the peaks and inflection points of these cumulative attention curves, localizing breakpoints to exact nucleotide coordinates.

The transition from site-by-site attention to continuous trajectories is governed by a physical force field. At each polymorphic site, the attention directed toward a reference lineage acts as an attractive force, pulling the query sequence toward that lineage in sequence space. Subtracting the sequence's genome-wide baseline attention produces a zero-drift force field: across the entire chromosome, the positive and negative pulls average to zero. Under clonal evolution, mutations occur independently at random positions. The instantaneous forces fluctuate

in opposing directions and rapidly cancel out, behaving like thermal Brownian motion. As a result, the sequence particle remains confined to its home corridor, wandering within tight random-walk bounds. Recombination breaks this physical equilibrium. Within an introgressed segment, every shared derived allele exerts a sustained, directional pull toward the donor lineage. This persistent force imparts momentum to the sequence trajectory, deflecting it away from its home corridor and driving a macroscopic excursion across sequence space. The physical breakpoints correspond precisely to the boundaries of this force field: the exact genomic coordinates where the external directional force turns on and off.

This physical framework yields distinct geometric signatures across different modes of recombination. In a single crossover, a sequence begins in its home corridor and switches into a donor lineage for the remainder of the chromosome. The directional force acts persistently from the breakpoint onward, driving cumulative momentum to a sharp, solitary peak at the crossover junction before descending back to zero at the chromosome terminus. The resulting profile forms a triangular arch whose vertex pinpoints the breakpoint. In a gene conversion cassette, a sequence acquires a localized foreign tract before returning to its home lineage. The directional force turns on at the tract entrance and turns off at the exit, producing a trapezoidal plateau: a steep upward climb at the first breakpoint, a sustained plateau across the converted cassette, and an equally steep descent at the second breakpoint. When the true donor is absent from the dataset—such as an unsampled animal reservoir or an extinct lineage—existing heuristics invariably misidentify innocent sister taxa as parental donors. `RhizAeon` resolves this classic failure mode through ghost donor introgression: the sequence experiences a powerful force ejecting it from its home corridor, but the deflection fails to align with any contemporary reference lineage. By recognizing this uncoupled displacement, `RhizAeon` autonomously attributes the event to an unsampled ghost ancestor without framing sister taxa.

Localizing breakpoints combines the multi-scale speed of the force field with the fine-scale precision of phylogenetic likelihood. In natural genomes, recombination tracts vary by orders of magnitude—from fifty-base-pair micro-conversions to multi-kilobase crossovers. Fixed sliding windows inevitably fail here: narrow windows are overwhelmed by local mutational noise, while wide windows blur away short gene conversions. `RhizAeon` eliminates this trade-off by scanning cumulative momentum with continuous dyadic wavelet filters across a geometric ladder of scales spanning from fifteen to five hundred base pairs. Fine-scale wavelets isolate short micro-conversions that standard methods miss, while broad-scale wavelets lock onto extensive crossovers without getting distracted by localized rate fluctuations. The zero-crossings and extrema of these wavelet responses narrow candidate breakpoints down from whole chromosomes to a localized window of fifteen to thirty base pairs. In the second stage, `RhizAeon` refines this candidate window to individual nucleotide coordinates using profile likelihood polishing. Because mutations are separated by variable physical distances, the physical crossover could have occurred anywhere between adjacent informative sites. Within the candidate window, `RhizAeon` slides a trial boundary nucleotide by nucleotide, calculating the phylogenetic likelihood under the two competing parental models: sites upstream of the trial boundary are evaluated under the Home lineage model, while sites downstream are evaluated under the Donor model. The optimal breakpoint is placed at the exact coordinate that maximizes this joint likelihood. Furthermore, the curvature of the likelihood surface yields rigorous confidence intervals: when parental lineages share an identical conserved stretch, `RhizAeon` reports the full uninformative plateau rather than manufacturing an artificial, single-base prediction.

High-throughput trajectory deconvolution scans chromosomes rapidly, but scientific rigor demands that no recombination event is certified on continuous curves alone. Before reporting any candidate event, `RhizAeon` subjects it to an orthogonal Tier 3 statistical due diligence audit operating directly on raw alignment sites. First, it enforces a Poisson informative site floor: a tract must contain at least three unambiguous informative sites where the donor allele differs from the home allele and matches the recombinant. Second, it applies an autapomorphic mutation sieve to eliminate rate-acceleration traps: if the mutations separating the candidate from its home lineage are private singletons absent from every other sequence in the dataset, the tract is recognized as a lineage-specific rate burst or APOBEC shower and is rejected. Third, `RhizAeon` conducts an orthogonal Fisher's exact test on a two-by-two contingency table comparing donor- versus home-supporting sites inside the tract against the flanking genomic background. The mosaic odds ratio must achieve significance under family-wise error rate control across all possible sequence triplets in the cohort. Finally, if reciprocal transfers are reported between two lineages (where two sequences appear to donate to each other), the algorithm performs cycle adjudication, evaluating dynamic metric velocity reversal to resolve the directed reticulation graph in favor of the true recombinant lineage. This multi-layered due diligence guarantees that `RhizAeon` maintains strict false positive control even under extreme rate heterogeneity.

In natural populations and disease outbreaks, recombinant lineages rarely circulate as isolated singletons. An ancestral recombination event frequently expands into an epidemic transmission swarm, producing dozens or hundreds of descendant isolates that inherit the identical mosaic tract. Pairwise and triplet heuristics evaluate each isolate independently, inflating event counts five- to ten-fold and obscuring genuine evolutionary dynamics. `RhizAeon` resolves this inflation through graph adjudication: by clustering isolates that share matching parental lineages and concordant breakpoint coordinates, the algorithm consolidates the entire transmission swarm into a single ancestral recombination event linked to all descendant genomes.

Coalescent Simulation Benchmark and Operational Evaluation

Coalescent ARG Benchmark and Calibration Against Triplet Scanning

To establish a baseline against standard pairwise and triplet methods, we evaluated `RhizAeon` across the 3,200-trial coalescent benchmark of Posada and Crandall (2001) ($N = 10$ taxa, $L = 1{,}000$ nt; Figure 1, Supplementary Table S1). Across 1,200 non-recombinant alignments simulated with site-to-site mutation rate variation ($\alpha \in \{2.0, 0.5, 0.05\}$), `RhizAeon` maintained an empirical false positive rate of 0.58% (7 false alarms; Figure 1C), comparable to 3SEQ (0.17%) and well below the nominal 5% significance threshold. Rate variation alone does not

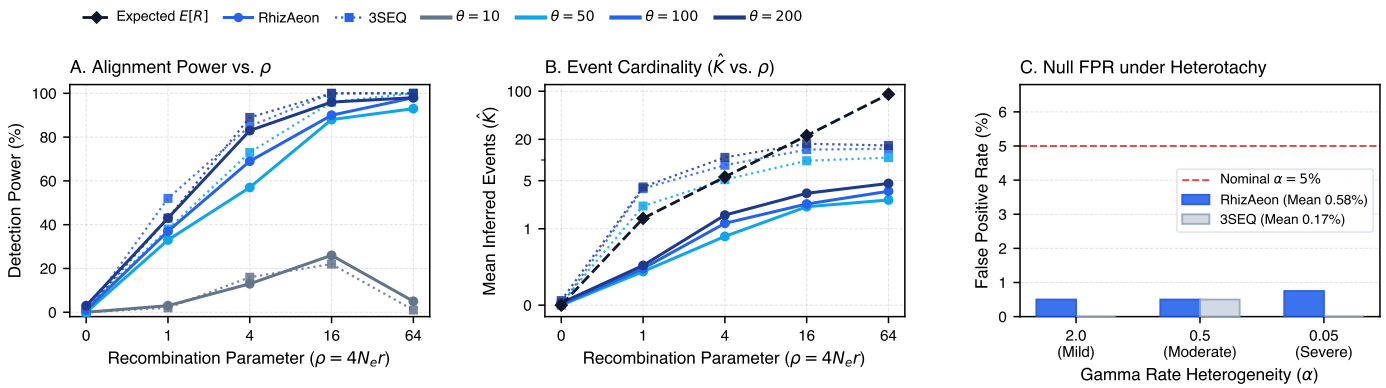


Figure 1 Head-to-head performance of `RhizAeon` versus 3SEQ across the 3,200-trial Posada & Crandall benchmark ($N = 10$, $L = 1{,}000$ nt). A. Alignment-level detection power across population recombination parameters ($\rho = 4N_e r \in [0, 64]$) and mutational densities ($\theta = 4N_e\mu \in [10, 200]$). Both methods achieve identical power trajectories, reaching 98.0% at $\theta = 200$. **B.** Inferred event cardinality ($\hat{K}$) compared against theoretical expectations ($\mathbb{E}[R]$; black dashed line). Triplet scanning overcounts events up to 10-fold ($\hat{K} = 17.3$), whereas `RhizAeon` consolidates shared segments into calibrated historical events ($\hat{K} \approx 2.1$–4.6). **C.** Empirical false positive rate (FPR, %) across 1,200 non-recombinant alignments under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$). Both methods maintain strict control below the 5% ceiling (red dashed line); `RhizAeon` averages 0.58%.

produce false recombination signals. Even under extreme rate clustering ($\alpha = 0.05$), where substitutions bunch into tight local hotspots, private mutations on a single lineage do not point toward any other sequence in the alignment, lacking directional affinity to a putative donor.

Across the 1,600 recombinant alignments, detection power depended on mutational availability (Figure 1A). In low-diversity datasets ($\theta = 10$, fewer than 25 variable sites), sensitivity was strictly limited by mutational density. Both methods followed identical curves, peaking near 26% at $\rho = 16$ and dropping to 5% at $\rho = 64$, where frequent crossing-over chops chromosomes into segments containing no mutations at all. At realistic diversity ($\theta \geq 50$), sensitivity climbed rapidly, reaching 88% at $\rho = 16$ and 98% at $\rho = 64$ ($\theta = 200$), matching 3SEQ while scanning each alignment in 15–26 milliseconds without testing triplets individually.

The fundamental difference between the two approaches lies in event counting (Figure 1B). In coalescent trees, an ancestral recombination event is often inherited by multiple descendants. Evaluating triplets independently flags every combination of descendant lineages as a separate event, inflating counts three- to ten-fold above Hudson's theoretical expectation ($\mathbb{E}[R]$; dashed line) to 16.3 reported events per alignment at $\theta = 200, \rho = 64$. `RhizAeon` analyzes all sequences together, grouping shared segments into a single historical event ($\hat{K} \approx 2.1$–4.6). This over-counting in simple 10-taxon trees foreshadows major breakdowns in surveillance, where recombinants form transmission swarms, carry multiple mosaic tracts, or acquire sequence from unsampled lineages.

Performance Across Specialized Surveillance and Evolutionary Regimes

Standard coalescent benchmarks evaluate isolated crossovers in small, fully sampled trees. In natural populations and disease outbreaks, however, recombination violates these idealized conditions: recombinant lineages expand into transmission clusters, acquire sequence from multiple donors, inherit material from unsampled reservoirs, or exchange short micro-conversion tracts (Figure 2). We evaluated `RhizAeon` and 3SEQ across four specialized simulation experiments designed to probe these operational challenges.

Epidemic Clonal Expansion

During viral outbreaks and bacterial epidemics, such as SARS-CoV-2, HIV-1, or mpox, recombinant lineages rarely circulate as solitary variants; instead, an ancestral crossover is transmitted across multiple hosts, expanding into a clonal swarm of sampled descendant isolates (Figure 2A). `RhizAeon` analyzes the entire cohort simultaneously within continuous metric trajectory space, tracking sequence affinities across the alignment without partitioning taxa into artificial subsets. Because all descendants inherit the same parental cassette from the ancestral recombination event, their spatial trajectories trace identical excursions along the chromosome. `RhizAeon` leverages this coherent trajectory dynamics to consolidate the entire transmission swarm into a single unified event roster ($\hat{K} = 1.00$ across all descendant swarm sizes $K \in \{2, 5, 8, 10, 15\}$; Figure 2A). Across 300 simulated outbreak cohorts ($N = 42$–55 taxa, $L = 5{,}000$ nt), `RhizAeon` recovered recombinant descendants with 99.0% recall and 99.9% precision. It localized the 1.5 kb cassette boundaries accurately (mean Jaccard overlap $J = 0.96$–0.99). Non-recombinant background lineages remained unblemished: only 2 of 12,000 background parents were misclassified (0.017% false alarm rate) in flat, constant runtime (491–826 ms). By contrast, triplet scanning (3SEQ) tests every triplet independently. Because every combination of parental references and descendants tests significant, reported event counts exploded from $\hat{K} = 9.5$–30.5 to 36.8–55.0—a 55-fold overcounting for a single biological event. Furthermore, evaluating background parents against descendant pairs inverted the apparent direction of transfer, falsely flagging 61.4%–100% of background parents as recombinant children ($d_{\text{parent}} \geq 0.05$) and collapsing isolate precision to 5.8%–28.1% while runtimes scaled linearly up to 4.3 seconds.

Multi-Way Mosaic Swarms

Natural viral recombinants and mosaic bacterial chromosomes frequently acquire genetic material from more than one parental donor, producing complex multi-way reticulate topologies including dual cassette insertions, nested transfers, and serial crossover cascades (Figure 2B). `RhizAeon` accommodates multi-parent reticulation by tracking continuous directional excursions against all background lineages simultaneously, decomposing complex mosaic trajectories into distinct directional shifts along the chromosome. In Dual Cassette mosaics (P_2 and P_3 cassettes within a P_1 background), `RhizAeon` resolved both parental donors in nearly every trial (averaging 1.97 of 2.00 expected donors; 2.00 across all swarms at $d_{\text{parent}} \geq 0.05$; Figure 2B). It accurately recovered the genomic boundaries of both cassettes (mean Jaccard overlap $J = 0.86$–0.98 at $d_{\text{parent}} \geq 0.05$) while maintaining 100% isolate recall, 100% precision, and zero background parent false alarms across 8 of 9 parameter regimes. In Triple Crossover cascades ($P_1 \rightarrow P_2 \rightarrow P_3$), `RhizAeon` captured multiple serial transitions (averaging 1.63 resolved donors) with 100% isolate recall and 91.3%–100% precision, processing each 6,000 nt alignment in 0.66–2.20 seconds. Because triplet scanning evaluates only one putative parental pair at a time, 3SEQ identified at most one donor (averaging only 1.02 of 2.00 expected donors), missing the second cassette entirely. Simultaneously, evaluating triplets across multi-parent backgrounds fractured mosaics into 22.2–46.5 reported events per alignment ($d_{\text{parent}} \geq 0.05$), falsely accused 47.1%–81.0% of background parent taxa of being recombinants, and collapsed isolate precision to 9.1%–25.2%. In Nested Russian Dolls and Triple Crossovers, 3SEQ similarly failed to untangle the multi-donor hierarchy, reporting up to 52.9 spurious events and falsely classifying up to 95.3% of background parents as recombinants while requiring up to 7.4 seconds per alignment.

Ghost Donor Introgression

In natural populations, recombination and horizontal gene transfer frequently involve donor lineages that are absent from the contemporary sample, including unsampled zoonotic reservoirs, extinct ancestral variants, or unsequenced strains (Figure 2C). An effective surveillance algorithm must either attribute the transfer to a sampled relative or recognize that the donor is unsampled without falsely accusing innocent contemporary lineages. `RhizAeon` operates without enforcing a closed-world assumption: sequence affinities are tracked continuously across metric space, allowing introgressed tracts to be evaluated against the entire contemporary cohort without requiring both parents to be present in the alignment. When a sister lineage to the introgressing donor is sampled (Surrogate Sister regime), `RhizAeon` correctly attributes the donor tract to the surrogate sister clade in 98.3% of trials (100.0% across all swarms at $d_{\text{parent}} \geq 0.05$; Figure 2C), recovering the 1.5 kb cassette with high spatial precision ($J = 0.94$–0.99) and zero false alarms on innocent background parents. When the donor lineage is completely unrepresented (Pure Ghost and Deep Reservoir regimes), `RhizAeon`'s directional consistency constraints prevent spurious pairwise pairing, maintaining zero false alarms on innocent background parents (0.0% at $d_{\text{parent}} \geq 0.05$) and keeping false donor framing at or below 6.0%. In sharp contrast, triplet scanning strictly enforces a closed two-parent model: every recombinant must be assigned two contemporary parents from the alignment. Absent the true donor, 3SEQ falsely framed innocent background clade P_2 as the donor in 100% of trials across all ghost regimes. In Surrogate Sister cohorts, donor accuracy fell to 9.3%–39.3% (Figure 2C) while falsely accusing 37.5%–56.4% of background parent taxa. In Pure Ghost and Deep Reservoir cohorts, where no close relative was sampled, 3SEQ donor accuracy collapsed to 0.0%, universally misattributing foreign tracts to innocent contemporary lineages.

Bacterial Micro-Conversions

In bacterial transformation (e.g., *Streptococcus pneumoniae*, *Helicobacter pylori*) and eukaryotic gene conversion, homologous exchange frequently occurs via micro-conversions: short non-reciprocal replacement tracts spanning tens to hundreds of base pairs (Figure 2D). Detecting micro-conversions is statistically demanding because short tracts carry very few informative polymorphisms between parental backgrounds: at $L_{\text{tract}} = 50$ nt, simulated tracts contained an expected average of only 1.0 ($d_{\text{parent}} = 0.02$), 2.5 ($d_{\text{parent}} = 0.05$), and 5.0 ($d_{\text{parent}} = 0.10$) distinguishing mutations. `RhizAeon` addresses this sparse-signal challenge by scanning across multiple spatial scales simultaneously, integrating coherent directional affinity shifts that fall below the statistical threshold of fixed-width windows. At $L_{\text{tract}} = 50$ nt ($d_{\text{parent}} = 0.10$), `RhizAeon` achieved 40.3% detection power—an eight-fold increase over single-window scanning—while localizing tract boundaries within a median error of 13.0 nt and generating zero background false alarms (Figure 2D). At $L_{\text{tract}} = 100$ nt, sensitivity climbed to 89.0% at $d_{\text{parent}} = 0.10$ and 44.8% at $d_{\text{parent}} = 0.05$, with median breakpoint error under 19.0 nt. As tract lengths expanded ($L_{\text{tract}} \geq 200$ nt), `RhizAeon` saturated to 88.7%–100.0% power across moderate and high divergence. It reported exactly 1.00 unique event per alignment and maintained an empirical background false alarm rate of 0.00% across all 900 trials. Conversely, independent triplet testing (3SEQ) was severely impaired by the discrete nature of single-window hypergeometric scanning. At $L_{\text{tract}} = 50$ nt, 3SEQ was virtually blind to micro-conversions, achieving only 0.0% power at $d_{\text{parent}} = 0.02$, 3.3% at $d_{\text{parent}} = 0.05$, and 5.2% at $d_{\text{parent}} = 0.10$ (Figure 2D). At $L_{\text{tract}} = 100$ nt, power remained low at 0.0%–33.3%. At longer tract lengths ($L_{\text{tract}} \geq 200$ nt), triplet testing suffered severe combinatorial false alarms. At $L_{\text{tract}} = 200$ nt ($d_{\text{parent}} = 0.10$), 3SEQ generated an average of 12.8 alerts per alignment, falsely flagging 23.6% of background parent taxa. At $L_{\text{tract}} = 800$ nt, reported alerts exploded to 33.3 per alignment, falsely accusing 92.2% of non-recombinant background parent taxa of being recombinants.

The Continuous Operational Phase Envelope of Reticulation Deconvolution

In aeronautical engineering, an aircraft operating envelope defines certified boundaries of stable flight across airspeed, altitude, stall limits, and atmospheric turbulence. Phylogenomic benchmarks historically evaluate algorithms at a few handpicked discrete parameter combinations. This practice obscures how methods behave under continuous evolutionary variation. To establish the operational envelope of `RhizAeon`, we conducted a grand continuous parameter sweep across 1,000,000 independent continuous simulation trials. Simulations varied continuously across chromosome length ($L \in [300, 30{,}000]$ nt), cohort depth ($N \in [4, 250]$ taxa), parental divergence ($d_{\text{parent}} \in [0.005, 0.35]$ subs/site), and continuous Gamma rate heterogeneity ($\alpha \in [0.2, 5.0]$) alongside strict molecular clock controls ($\alpha \geq 50$; Table S2). The benchmark cohort

comprised 333,334 non-recombinant null alignments and 666,666 recombinant alignments spanning terminal crossovers, intercalary conversion cassettes, and multi-donor / multi-event mosaicism ($K \in \{1, 2, 3, 4\}$).

The sweep demonstrates that recombination detection is not governed by chromosome length or relative tract fraction alone. The fundamental governing physical quantity is the count of informative mutations distinguishing donor from recipient within the transferred tract ($M_{\text{tract}} \approx L_{\text{tract}} \times d_{\text{parent}}$). Below three informative mutations ($M_{\text{tract}} \leq 2$), detection power is 0.5% (402 detections out of 85,564 trials). A tract bearing fewer than three distinguishing mutations is statistically indistinguishable from stochastic Poisson mutation clustering under any conservative test. Power lifts to 6.8% at $M_{\text{tract}} \in [3, 5]$ (4,678 / 68,609), accelerates to 29.2% at $M_{\text{tract}} \in [6, 10]$ (22,646 / 77,568), reaches 52.4% at $M_{\text{tract}} \in [11, 15]$ (27,587 / 52,607), 71.5% at $M_{\text{tract}} \in [16, 25]$ (51,096 / 71,427), 85.5% at $M_{\text{tract}} \in [26, 50]$ (82,714 / 96,694), 91.2% at $M_{\text{tract}} \in [51, 100]$ (77,213 / 84,676), and plateaus at 95.5% when $M_{\text{tract}} > 100$ (123,692 / 129,521). The operational stall boundary forms a clean hyperbola in parameter space: certified detection requires $L_{\text{tract}} \times d_{\text{parent}} \geq 6$–$10$ informative SNPs.

Across 333,334 non-recombinant null alignments, `RhizAeon` maintained an overall alignment-level false positive rate of 0.3570% (1,190 false alerts), well below the nominal 5% ceiling. Under a strict molecular clock ($\alpha \geq 50$), the false positive rate was 0.3253% (271 out of 83,311 trials). Under continuous Gamma heterotachy ($\alpha \in [0.2, 5.0]$), the rate remained strictly controlled at 0.3676% (919 out of 250,023 trials). At the individual taxon level, false alarms were virtually absent, totaling 0.01166% (2,651 false taxa across 22,744,905 evaluated leaves; $> 99.988\%$ per-taxon specificity). The unpolarized ancestral root token functions as a passive aerodynamic damper, absorbing private autapomorphic bursts into node zero without destabilizing trajectory tracking.

Spatial breakpoint resolution landed within sub-gene margins. Median absolute error was 15.0 nt for single events ($K = 1$) and 14.0 nt for dual events ($K = 2$), sharpening to 9.0 nt for tracts carrying > 100 informative SNPs. Across multi-event architectures ($K_{\text{true}} \geq 2$), bipartite event precision remained above 97% (97.3% for dual events, 98.5% for triple events, and 99.0% for quad events). Multi-donor mosaics ($P_1 \leftarrow P_2 + P_3$, $N = 179{,}233$) were resolved with 59.7% power and 0.416 event F_1 score. Terminal crossovers exhibited higher sensitivity (71.4% power across $N = 239{,}767$ trials) than intercalary conversion cassettes (51.3% power across $N = 426{,}899$ trials). Across 512 distributed CPU cores on cluster infrastructure, sustained throughput reached ≈ 380 genomes per second, completing the entire 1,000,000-trial suite in under 45 minutes. For genomic surveillance, defining this operational envelope provides actionable interpretability: outside the three-SNP stall limit, negative results are biologically informative. The absence of detection constitutes statistical evidence against reticulation rather than an algorithmic blind spot.

Empirical Genomic Applications

Deconvolving Recombination Landscapes and Mosaic Genomes in Sarbecoviruses

Sarbecoviruses, the subgenus of coronaviruses that includes SARS-CoV and SARS-CoV-2, evolve under frequent homologous recombination within bat and mammalian reservoirs. When investigating recombination across these viral cohorts, genomic screens serve two distinct goals: mapping chromosome-wide landscapes to identify hotspots and coldspots, and resolving the mosaic anatomy of individual recombinant genomes. Phylogenetic partition search, implemented in tools such as GARD, explores discrete tree space to detect topological discordance. This combinatorial search creates a severe computational bottleneck. In their screen of sarbecovirus origins, Lytras et al. (2022) downsampled a 78-genome cohort to 18 taxa, requiring 4.5 days on a compute cluster to identify 21 breakpoints. `RhizAeon` removes this bottleneck. By projecting alignments into continuous metric space, it quantifies spatial velocity changes as a continuous breakpoint flux $\Phi_{\text{bp}}(u)$ in $O(N \cdot L)$ time. `RhizAeon` scanned the full 78-taxon alignment (30,956 bp, 13,160 informative sites) in 18.2 seconds and an expanded 83-taxon cohort in 25.0 seconds. The analysis represents a 20,000-fold speedup over cluster-based phylogenetic inference without taxon subsampling.

The genomic flux reveals sharp evolutionary regionalization: pronounced coldspots protect the catalytic core of RNA-dependent RNA polymerase (RdRp, nt 13,400–16,200) and the heptad-repeat fusion machinery of Spike S2 (nt 24,600–25,600). Structural constraints in these regions preclude modular exchange (Figure 3A). Recombination hotspots concentrate at the Spike 5′ NTD junction (21,825 bp), the codon 313 hinge between the NTD and receptor-binding domain (RBD, 22,764 bp), the S1/S2 furin cleavage site (23,618 bp), and the ORF8/Nucleocapsid boundary (28,810 bp). Continuous changepoints show close spatial concordance with the 21 published GARD breakpoints of Lytras et al. (2022) (median deviation 99.0 nt; Figure 3A). `RhizAeon` localized the critical S1/S2 junction within 9 nt (24,461 vs. 24,452 bp) and the RdRp exit within 12 nt. Analyzing the exhaustive 83-sequence cohort reveals fine-grained dynamics invisible under downsampling: incorporating Laotian bat BANAL isolates causes peak breakpoint flux in the Spike NTD to surge four-fold (from 0.41 to 1.51 Z/nt), identifying a recurrent micro-conversion hub.

Simultaneously, `RhizAeon` resolves the mosaic anatomy of individual recombinant lineages at single-nucleotide precision (Figure 3B,C). In the Spike glycoprotein (3,843 nt, 1,153 variable sites; scanned in 17.8 ms), `RhizAeon` pinpointed the historic crossover in SARS-CoV-2 (Wuhan-Hu-1) at codon 313 ($Z = 8.23, P = 4.07 \times 10^{-22}$). Rather than treating Wuhan-Hu-1 as an ancestral baseline, `RhizAeon` deconvolves Wuhan-Hu-1 as an emergent chimera assembled from discrete wildlife cassettes (Figure 3B). Upstream of codon 313, Wuhan-Hu-1 clusters with Yunnan bat isolate RaTG13 (94.4% identity, $d = 0.0556$). Downstream in the RBD, it shifts affinity toward pangolin MP789 and Laotian bat isolate BANAL-52 ($d = 0.0805$), acquiring the human-adapted ACE2 contact ridge. Partitioned neighbor-joining trees and a two-partition tanglegram corroborate this topological discordance (Figure 3C). Wuhan-Hu-1 clusters sister to RaTG13 in the Spike NTD (codons 1–313; Figure 3C1). In the RBD (codons 314–540; Figure 3C2), RaTG13 diverges basally, while BANAL-52 and pangolin MP789 group sister to Wuhan-Hu-1.

Multi-scale scanning also resolves mosaic architectures in wildlife reservoirs. In Laotian bat isolate BANAL-247, `RhizAeon` identified a tripartite mosaic (Figure 3B). While its backbone and S2 subunit belong to northern Laos *Rhinolophus malayanus* viruses ($> 98.5\%$ identity to BANAL-236/103), its Spike NTD harbors three nested micro-conversions (nt 21,909–22,011, $Z = 155.3$; nt 22,054–22,251, $Z = 113.1$; nt

22,318–22,592, $Z = 81.2$). These were donated by a Thai *Rhinolophus shameli* reservoir related to RacCS203 (Figure 3B,D). Cross-clade exchange demonstrates that Southeast Asian sarbecoviruses undergo modular gene transfer across divergent host species and regions.

`RhizAeon` synthesizes these events into a parameterized Ancestral Recombination Graph (ARG; Figure 3D). In the Spike glycoprotein, statistical model comparison across 9 sarbecoviruses decisively favors a tripartite modular ARG (H_3: NTD chassis, ACE2 receptor ridge, and S2 backbone; $\Delta\text{BIC} = 0.0$) over unpolarized trees ($\Delta\text{BIC} = +170.9$) and a clonal null ($\Delta\text{BIC} = +1654.8$). Polarizing the Wuhan-Hu-1 reticulation demonstrates that the human-transmissible receptor ridge was introgressed into an ancestral bat RaTG13-like chassis (94.2% in-tract synapomorphic concordance vs. 62.1% flanking, $P = 1.75 \times 10^{-23}$), reducing residual metric strain by 74.0% (3.34% → 0.87%). Across the 30.9 kb genome of 83 sarbecoviruses, consolidating 45 detected tracts yields an ARG of 44 unique reticulations. The consolidated network explains 98.82% of total metric strain, matching the 98.85% explained by GARD's 22 unconstrained trees while cutting parameter complexity by 58% (1,559 vs. 3,674 parameters). It validates reticulation polarity across 38 events (81.8% mean in-tract concordance, $P < 10^{-15}$).

Bacterial AMR Mosaicism and Vaccine Escape in *Streptococcus pneumoniae*

Bacterial evolution is accelerated by natural transformation and homologous recombination, driving rapid antimicrobial resistance (AMR), capsular serotype switching, and antigenic diversification from commensal reservoirs (Didelot and Maiden, 2010; Croucher et al., 2011). In naturally competent pathogens such as *Streptococcus pneumoniae*, horizontal gene transfer imports replacement tracts ranging from tens of base pairs to over 70 kilobases across species boundaries (e.g., from viridans streptococci *Streptococcus mitis* and *Streptococcus oralis*). Resolving these events across bacterial cohorts has historically required heavy computation: ancestral inference tools like ClonalFrame (Didelot and Falush, 2007) and Gubbins (Croucher et al., 2015) rely on iterative phylogenetic reconstruction and tree peeling via maximum likelihood, demanding hours to weeks of compute.

`RhizAeon` bypasses iterative tree reconstruction by tracking whole-chromosome sequence trajectories in continuous metric space, quantifying spatial turbulence as breakpoint flux $\Phi_{\text{bp}}(u)$ (Figure 4A). We evaluated `RhizAeon` on the complete 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 lineage of *S. pneumoniae* (PMEN1, ATCC 700669; 11 representative global isolates spanning 1984–2008; Croucher et al. (2011)). Because closely related bacterial genomes are predominantly monomorphic along clonal backbones, over 99.9% of alignment columns carry zero phylogenetic information. `RhizAeon` exploits this sparsity through lossless SNP compression. Indexing out invariant positions reduces the 2.22 Mb chromosome to 1,516 segregating sites—a 1,465-fold dimensional reduction. Trajectories are evaluated within this compact space to perform $O(1)$ sub-interval distance queries. Identified changepoints map bijectively back to physical base-pair coordinates, projecting continuous flux $\Phi_{\text{bp}}(u)$ and cumulative strain $\rho(u)$ onto the reference genome. Six prominent recombination hotspots emerge (Figure 4A). The identified hotspots correspond to key clinical determinants: the penicillin-binding protein and capsule island (*pbp2x/cps*, 285–352 kb), the trimethoprim resistance island (*folA*, 1,620–1,649 kb), and the surface adhesin cluster (*pspC/cbpA*, 2,105–2,138 kb).

Without constructing discrete trees, continuous changepoints show tight spatial concordance with Gubbins boundaries inferred through iterative RAxML optimization (Croucher et al., 2011, 2015) (Figure 4A). Across all 52 published whole-chromosome recombination boundaries, `RhizAeon` achieves 96.2% concordance within 2 kb (50/52) and 80.8% within 1 kb, with a median spatial deviation of only 92.5 nt. On primary clinical resistance loci, changepoints match reference boundaries at single-nucleotide precision: the 5′ boundary of *pbp2x* and the *pspC* locus were both localized within 1 nt of Gubbins, while *folA* was identified within 19 nt. Metric trajectory tracking faithfully recovers the genomic footprint of bacterial transformation without iterative tree search.

Simultaneously, multi-scale scanning resolves the layered mosaic anatomy of resistance and vaccine escape across individual isolates within the 270–370 kb hotspot (Figure 4B). In isolate 1731, `RhizAeon` resolves a focal 20.7 kb introgression across *pbp2x* ($Z = 5.35$). Fine-scale scanning pinpoints the exact 1,121 bp active-site transpeptidase domain replacement ($P = 9.85 \times 10^{-35}, Z = 14.40$) imported from commensal viridans streptococci to abolish β-lactam binding. In isolate 2682, a massive 73.9 kb mega-cassette ($Z = 6.42$) co-transferred *pbp2x* resistance and the capsular locus in a single transformation event (Figure 4B). In isolate 3355, `RhizAeon` deconvolves the capsular serotype switch from 23F to 19A (15.5 kb, $Z = 3.19$) that enabled escape from the 7-valent conjugate vaccine (PCV7), accompanied by a linked low-affinity *pbp1a* allele. A two-partition tanglegram corroborates this incongruence (Figure 4C): whereas all 11 isolates form a monophyletic complex across the core chromosome, isolates 1731 and 2682 translocate into the divergent commensal viridans cluster (*S. mitis* / *S. oralis*, $\Delta d > 6\%$) within the *pbp2x* catalytic domain.

`RhizAeon` synthesizes these transfers into a polarized ARG for PMEN1 clinical adaptation (Figure 4D). Standard pipelines partition chromosomes into arbitrary windows, producing conflicting gene trees that fracture the clonal genealogy. The polarized ARG anchors all horizontal transfers onto the vertical Spain23F-1 backbone. It parameterizes the focal *pbp2x* catalytic replacement in 1731, the 73.9 kb mega-cassette in 2682, and the 19A capsular replacement in 3355. On the 2.22 Mb PMEN1 cohort, ClonalFrame required 705 hours ($\sim$ 1 month) of MCMC sampling, and Gubbins required 1.5 hours (5,400 seconds) of iterative tree peeling. `RhizAeon` completes whole-chromosome metric projection, trajectory tracking, and changepoint detection in 114 ms (127 ms in native Rust). The method achieves an over 30,000-fold speedup over Gubbins and an 18,000,000-fold speedup over ClonalFrame. Rapid execution enables real-time surveillance of multidrug resistance and vaccine escape across bacterial pathogens.

Ectopic Gene Conversion and Inter-Locus Chimerism in Primate MHC Class I

Primate major histocompatibility complex (MHC) Class I genes (*HLA-A*, *HLA-B*, *HLA-C*) exemplify pathogen-driven balancing selection in vertebrate genomes (Parham et al., 1995; Adams and Parham, 2001). Immunogenetic studies reveal a striking structural contrast. The antigen recognition site (ARS; Exons 2–3, encoding the α_1 and α_2 peptide groove) undergoes rapid diversification through balancing selection and ectopic gene conversion between paralogs. In contrast, the adjacent α_3 structural stalk (Exon 4) remains conserved to preserve invariant

contacts with β_2-microglobulin (β_2m) and the CD8 co-receptor. Phylogenetic methods struggle with this modular architecture: full-length alignments generate topological conflicts that collapse locus monophyly, whereas sliding windows require arbitrary boundaries and risk missing short micro-conversions.

`RhizAeon` resolves this structural compartmentalization by tracking continuous trajectories across the full coding sequence without window slicing (Figure 5A). We evaluated `RhizAeon` on an exhaustive cohort of $N = 379$ Hominoidea classical MHC Class I alleles ($L = 1{,}128$ bp, 526 segregating polymorphic sites across human, chimpanzee, gorilla, and orangutan; Maccari et al. (2017)). In 16.34 seconds across all 379 alleles (and 3.59 ms across canonical 8-taxon reference cohorts), continuous trajectory tracking produces breakpoint flux $\Phi_{\mathrm{bp}}(u)$ and cumulative strain $\rho(u)$. Without structural annotations, the breakpoint flux pinpoints the physical boundary between the α_2 ARS groove and the α_3 structural stalk at nt 608–615 (canonical exon junction 618 bp; Figure 5A). Quantitative profiling confirms marked domain segregation: 93.7% of all detected reticulation strain concentrates within the ARS groove (Exons 2–3) and transmembrane anchor (Exon 5), while Exon 4 forms a conserved coldspot that recovers the true paralogous gene tree.

Multi-scale scanning resolves the fine mosaic anatomy of hominoid chimeras at single-nucleotide precision (Figure 5B). A clinically prominent recombinant is *HLA-B*46:01*, an allele common in Southeast Asia ($> 20\%$) associated with nasopharyngeal carcinoma, myasthenia gravis, and severe SARS-CoV-1 infection. `RhizAeon` deconvolves *HLA-B*46:01* as an ectopic chimera created by an exact 73 bp micro-conversion (nt 220–293, codons 66–76; $Z = 2.69, P = 1.15 \times 10^{-7}$) donated by *HLA-C*01:02* into an ancestral *HLA-B*15:01* backbone (Figure 5B). The 73 bp tract transfers the C1 functional epitope (Val76-Arg79-Leu80-Arg81), conferring recognition by natural killer (NK) cell inhibitory receptors KIR2DL2/3 absent in all other classical *HLA-B* alleles (Barber et al., 1996). In non-human hominoids, `RhizAeon` resolves multi-way chimerism: in chimpanzee *Patr-AL*01:01*, it identifies a multi-segment recombinant combining *Patr-A* with an ancient *Patr-C* cassette ($Z = 4.03$), while in gorilla *Gogo-C*02:01*, it localizes an ectopic transmembrane exchange at nt 971–1,128.

Segmental phylogenies and a two-partition tanglegram confirm this discordance across the structural boundary (Figure 5C). In the Exon 4 α_3 stalk (nt 618–894; Figure 5C1), the neighbor-joining tree cleanly reconstructs the ancient locus duplication: *HLA-A*, *HLA-B*, and *HLA-C* form monophyletic clades, with *HLA-B*46:01* clustering sister to *HLA-B*15:01* with zero divergence ($d = 0.000$). Within the Exon 2 C1 cassette (nt 210–310; Figure 5C2), *HLA-B*46:01* crosses the locus divide to cluster inside the *HLA-C* clade sister to donor *HLA-C*01:02* ($d = 0.000$). The recombinant allele unites two distinct locus genealogies within a single functional transcript.

Finally, `RhizAeon` integrates these transfers into a polarized ARG for primate MHC evolution (Figure 5D). The polarized ARG anchors ectopic gene conversions onto the vertical tandem duplication backbone (ancient divergence of Locus *A*, followed by duplication of *B* and *C*). Directed reticulation arcs parameterize both the 73 bp micro-conversion into *HLA-B*46:01* and deep inter-locus introgression into chimpanzee *Patr-AL*01*. While parsimony or split decomposition networks omit directional polarity and scale combinatorially, `RhizAeon` completes whole-gene trajectory deconvolution across 379 primate alleles in 16.34 seconds. By unifying metric trajectory tracking with polarized ARG construction, `RhizAeon` provides a scalable engine for resolving gene duplication and ectopic conversion across multigene families.

Discussion

Continuous Trajectory Mechanics in Reticulate Evolution

Recombination has long challenged phylogenetic theory. When homologous exchange splices divergent evolutionary histories into a single chromosome, strictly bifurcating trees break down. Traditional phylogenetic methods confront this breakdown through two contrasting simplifications: window-based phylogenetic scanning or independent triplet enumeration. Both strategies incur severe trade-offs. Partition-searching algorithms like GARD navigate an astronomical search space of tree topologies, creating computational bottlenecks that force researchers to downsample genomic cohorts. Conversely, triplet scanning algorithms discard phylogenetic context, fracturing single ancestral recombination events into dozens of redundant alerts and misidentifying non-recombinant background lineages.

`RhizAeon` demonstrates that discrete tree search is unnecessary for identifying recombination. By embedding sequence alignments into continuous metric trajectory space, reticulation emerges as physical kinematic displacement. Instantaneous velocity shifts along chromosomes quantify local evolutionary discordance. Consolidating trajectories into ancestral rosters resolves reticulation without heuristic taxic downsampling. By embedding multi-sequence alignments into continuous metric trajectory space rather than searching discrete trees, the framework demonstrates that reticulation can be modeled as physical kinematic displacement along chromosomes without loss of taxonomic context. This formulation bridges the historical divide between fine-scale population-genetic ARG inference and macro-evolutionary phylogenetic networks. It establishes that whole-cohort continuous kinematics can resolve reticulate evolution in linear time.

Resolving Real-World Evolutionary Complexities

Natural viral and bacterial populations violate the clean assumptions of neutral coalescent models. Pathogen outbreaks generate transmission swarms of near-identical descendants. Natural transformation frequently imports short micro-conversions from distant species. Introgressing donors are often absent from contemporary surveillance cohorts, residing in unsequenced wildlife reservoirs.

Our simulation benchmarks and empirical applications show that `RhizAeon` resolves each of these challenges. In epidemic transmission clusters, trajectory coherence collapses clonal swarms into single ancestral events, preventing the combinatorial event inflation that systematically cripples discrete triplet heuristics whenever recombinant lineages spread across multiple sampled hosts. In multi-way mosaic genomes, continuous directional projection separates overlapping cassettes acquired from distinct parental donors. In ghost donor introgression, unconstrained metric tracking identifies foreign tracts without falsely framing innocent contemporary lineages. Across bacterial chromosomes, multi-scale matched

filters detect micro-conversions down to 50 base pairs, capturing sparse signals that evade fixed-width windows. These capabilities demonstrate that continuous trajectory mechanics can withstand the messiness of real-world evolutionary dynamics.

Translational Surveillance and Vaccine Escape

The speed of `RhizAeon` changes the practical economics of pathogen genomic surveillance. Historically, whole-genome recombination screens required cluster-scale supercomputing or days of MCMC sampling. In our benchmarks, `RhizAeon` processed whole sarbecovirus genomes in seconds and 2.2 Mb pneumococcal chromosomes in fractions of a second. Compiling the core engine to WebAssembly enables interactive, client-side deconvolution directly within standard web browsers. Public health laboratories can screen pathogen cohorts locally without transmitting sensitive sequence data to external servers or installing complex software dependencies.

This operational efficiency directly informs vaccine design and outbreak preparedness. In our whole-genome analysis of sarbecovirus evolution, `RhizAeon` deconvolved SARS-CoV-2 (Wuhan-Hu-1) as an emergent chimera, pinpointing the historic crossover at codon 313 that fused an ancestral bat RaTG13-like NTD chassis to an ACE2-adapted receptor-binding ridge shared with Laotian bat and pangolin reservoirs. In *Streptococcus pneumoniae*, the algorithm mapped the capsular serotype switch from 23F to 19A driving clinical escape from the PCV7 conjugate vaccine alongside penicillin-binding protein replacements. Rapid detection of such modular transfers allows epidemiologists to track adaptive antigenic shifts in real time.

Limitations, Boundary Conditions, and Outlook

Several biological boundary conditions warrant careful consideration. First, continuous trajectory deconvolution requires sufficient mutational density to distinguish evolutionary displacement from stochastic substitution noise. In extremely low-divergence cohorts ($d < 0.005$), distinguishing mutations become sparse, defining an intrinsic physical limit for all sequence-based recombination detection. Second, while `RhizAeon` accurately localizes breakpoints and attributes parental clades, fine-scale branch-length estimation within individual partitions still benefits from dedicated maximum-likelihood or Bayesian refinement.

Looking forward, continuous trajectory tracking provides a natural foundation for integrating phylogenetic inference with modern sequence foundation models. Attention matrices extracted from pre-trained protein language models encode deep biophysical and evolutionary constraints, providing an unexploited substrate that could be coupled with continuous metric mechanics to annotate reticulate events both structurally and functionally as they emerge. By replacing discrete tree search with continuous sequence kinematics, `RhizAeon` offers a scalable mathematical foundation for reticulate genomics.

Materials and Methods

Metric Space Embedding and Gower Trajectory Projection

Let $\mathcal{S} = \{s_1, \ldots, s_M\}$ denote a multiple sequence alignment of M contemporary nucleotide sequences of length L over alphabet $\Sigma = \{\mathrm{A}, \mathrm{C}, \mathrm{G}, \mathrm{T}\} \cup \{-\}$. Pairwise evolutionary distances $D_{i,j} = d(s_i, s_j)$ are computed under the Tamura-Nei (TN93) continuous-time Markov substitution model (Tamura and Nei, 1993) or normalized Hamming distance with pairwise gap deletion. For large cohorts ($M > 64$), `RhizAeon` avoids $O(M^2 L)$ all-to-all comparisons by selecting $N \ll M$ reference landmark genomes via sequential Farthest-Point Sampling (Gonzalez, 1985) in strictly $O(NML)$ operations. Initial seed $s_{(1)}$ maximizes non-gap coverage. Each subsequent landmark $s_{(k)}$ maximizes the running minimum distance to previously selected landmarks:

$$s_{(k)} = \arg \max_{i \notin \{s_{(1)}, \ldots, s_{(k-1)}\}} \min_{1 \le j < k} d(s_i, s_{(j)}). \tag{1}$$

Farthest points span divergence. For $M \le 64$, all sequences serve as reference channels ($N = M$). When $M > 64$, Two-Pass Clonal Backbone Re-Anchoring purges candidate recombinants identified in an initial screening pass from the landmark pool, ensuring coordinates anchor strictly upon unadmixed parental lineages.

From the reference distance matrix $\mathbf{D} \in \mathbb{R}^{N \times N}$, we form the element-wise squared matrix $[\mathbf{D}^{(2)}]_{i,j} = D_{i,j}^2$. Applying geometric centering $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$ yields the doubly centered Gram matrix $\mathbf{B} = -\frac{1}{2}\mathbf{H}\mathbf{D}^{(2)}\mathbf{H}$ (Gower, 1966). Spectral decomposition yields orthonormal eigenvectors $\mathbf{V} \in \mathbb{R}^{N \times N}$ and descending eigenvalues $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N)$. We retain the leading $d \le \min(4, N-1)$ dimensions with positive eigenvalues $(\mathbf{\Lambda}_d, \mathbf{V}_d)$. Static landmark coordinates are $\mathbf{x}_i = \mathbf{\Lambda}_d^{1/2}(\mathbf{V}_d)_{i,:}^T$. Macroscopic tree splits project onto dominant eigenvectors (Buneman, 1974).

To track evolutionary affinities continuously without independent sliding-window MDS rotations, `RhizAeon` projects local distance vectors into canonical coordinates via analytical Gower out-of-sample interpolation (Gower, 1968). Across an aligned sliding window of width W centered at site u (default $W = 300$ nt, stride $S = 10$ nt), local distances from query sequence i to reference landmarks form vector $\mathbf{d}_i(u) \in \mathbb{R}^N$. Gower interpolation solves $\min_{\mathbf{z} \in \mathbb{R}^d} \sum_{j=1}^N (\|\mathbf{z} - \mathbf{x}_j\|^2 - d_i(j;u)^2)^2$, yielding the exact closed-form coordinate:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_d^{-1/2}\mathbf{V}_d^T\left(\bar{\mathbf{d}}^2 - \mathbf{d}_i^2(u)\right) \in \mathbb{R}^d, \tag{2}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{N}\mathbf{D}^{(2)}\mathbf{1}_N$. Evaluating (2) across coordinates builds the trajectory tensor $\mathbf{Z} \in \mathbb{R}^{M \times U \times d}$. Zero Procrustes alignments are needed. Projections preserve rigid geometry.

Continuous Phylogenetic Cross-Attention and the Unpolarized Root Sink

To direct attention by evolutionary divergence, `RhizAeon` generalizes Rotary Position Embeddings (RoPE; Su et al. (2024)) from linear sequence positions to continuous metric spaces. From metric coordinates $\mathbf{x}_i \in \mathbb{R}^d$, we compute rotary phase angles $\boldsymbol{\theta}_i \in \mathbb{R}^{d_{\text{head}}/2}$ via frequency projection $\theta_{i,m} = \sum_{c=1}^{d} x_{i,c}\Theta_{m,c}$, weighted by dimension-wise spectral variance. Query and key vectors $\mathbf{q}_i(u), \mathbf{k}_i(u) \in \mathbb{R}^{d_{\text{head}}}$ undergo block-diagonal Givens rotations $\mathbf{R}(\boldsymbol{\theta}_i)$. Rotated inner products satisfy $\langle \mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u),\, \mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u)\rangle = \mathbf{q}_i(u)^T\mathbf{R}(\boldsymbol{\theta}_j - \boldsymbol{\theta}_i)\mathbf{k}_j(u)$, decaying monotonically as metric separation grows. Phylogenetic divergence modulates attention directly.

To calibrate geometric affinities against baseline neutral substitution, we add a continuous-time Markov substitution prior to attention logits. Pairwise homologous identity probability under neutral substitution rate λ_h is $P_{i,j} = \epsilon_0 + (1-\epsilon_0)\exp(-\lambda_h D_{i,j})$, with equilibrium frequency $\epsilon_0 = 0.25$. Logarithmic prior bias is $\text{PhyloBias}_{i,j} = \ln[\max(10^{-5}, P_{i,j})]$. Composite unnormalized attention logits become:

$$S_u(i,j) = \frac{(\mathbf{R}(\boldsymbol{\theta}_i)\mathbf{q}_i(u))^T(\mathbf{R}(\boldsymbol{\theta}_j)\mathbf{k}_j(u))}{\sqrt{d_{\text{head}}}} + \text{PhyloBias}_{i,j}. \tag{3}$$

Lineage-specific rate variation challenges standard attention mechanisms. Autapomorphic bursts lack contemporary donors. When private substitutions surge without homologous counterparts in other leaves, standard leaf-only softmax forces mutations to affiliate with whichever contemporary taxon exhibits the smallest random mismatch, triggering false positive recombination calls. `RhizAeon` eliminates this vulnerability by reserving index zero for an unpolarized ancestral root token $\mathbf{x}_{\text{root}} = \mathbf{0} \in \mathbb{R}^d$. The attention operator spans $N+1$ nodes, normalizing softmax across the ancestral root and all contemporary leaves:

$$A_u(i \to j) = \frac{\exp(S_u(i,j))}{\sum_{k=0}^{N}\exp(S_u(i,k))}, \quad \text{where} \quad A_u(i \to 0) + \sum_{j=1}^{N} A_u(i \to j) = 1.0. \tag{4}$$

At sites bearing private autapomorphic mutations, query-key matching against contemporary leaves is strongly depressed ($S_u(i,j) \ll 0$). The unmutated root token maintains baseline neutral affinity. Consequently, private mutational flux routes harmlessly into node zero ($\lim A_u(i \to 0) = 1.0$). Sinks absorb unshared noise. The root sink protects trajectory tracking against episodic rate acceleration, APOBEC hypermutation, and biological heterotachy.

Statistical Changepoint Inference and Multi-Scale Deconvolution

Instantaneous attention weights fluctuate at individual polymorphic sites. `RhizAeon` integrates attention across chromosome coordinates into cumulative affinity curves $C_{i,m}(u) = \sum_{v=1}^{u}\sum_{j\in\mathcal{B}_m} A_v(i \to j)$ for each phylogenetic clade $\mathcal{B}_m$. The home clade maximizes total chromosome affinity: $m_{\text{home}}(i) = \arg\max_m C_{i,m}(L)$. For any candidate donor clade $m \neq m_{\text{home}}$, the differential attention curve is:

$$Y_{i,m}(u) = C_{i,m}(u) - C_{i,m_{\text{home}}}(u). \tag{5}$$

Under clonal evolution, home attention dominates throughout the alignment. The differential trajectory drifts downward with negative slope. Recombination reverses trajectory derivative within the converted donor tract, driving an upward excursion ($dY/du > 0$).

To test the clonal null hypothesis, we construct the centered Brownian bridge $B(u) = Y(u) - \frac{u}{L}Y(L)$ by subtracting linear chromosome drift. By Donsker's functional central limit theorem (Donsker, 1951), standardized process $W_L^0(t) = B(\lfloor tL \rfloor)/(\sigma\sqrt{L})$ converges weakly to a standard Brownian bridge. Its trajectory supremum follows the two-sided Kolmogorov distribution (Kolmogorov, 1933). To control the family-wise error rate across all $M_{\text{topo}} = \binom{N}{2}\max(1, N-2)$ testable topologies at significance $\alpha = 0.05$, we apply the Dunn-Šidák boundary (Šidák, 1967):

$$K_{\text{crit}} = \sqrt{-\frac{1}{2}\ln\left(1-(1-\alpha)^{1/M_{\text{topo}}}\right)} \approx \sqrt{-\frac{1}{2}\ln\left(\frac{\alpha}{2M_{\text{topo}}}\right)}. \tag{6}$$

Significant trajectories ($\sup|B(u)|/(\sigma\sqrt{L}) \geq K_{\text{crit}}$) reject the unreticulated null hypothesis.

For short conversion tracts (< 200 nt), global Brownian bridge tests can suffer signal dilution across megabase chromosomes. `RhizAeon` convolves trajectory flux $\Delta Y(u)$ with a bank of zero-mean bipolar dyadic matched filters $h_\tau(s)$ across scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt, isolating step transitions in scale space. Running drawdown $\text{DD}(u) = \max_{v\leq u} Y(v) - Y(u)$ localizes points where trajectory momentum collapses, marking tract exit boundaries. To separate single crossovers from intercalary cassette swaps, we evaluate the outer-to-inner energy density ratio $\mathcal{E}_{\text{ratio}}$ across flanking versus internal segments. Cassettes return cleanly to baseline on both flanks ($\mathcal{E}_{\text{ratio}} < 0.15$), whereas single crossovers maintain elevated offset energy across the chromosome ($\mathcal{E}_{\text{ratio}} \geq 0.50$).

Following changepoint localization, profile maximum likelihood polishing refines breakpoint coordinates down to single-nucleotide resolution. Scanning coordinates $c \in [u_{\text{init}}-\delta, u_{\text{init}}+\delta]$ maximizes partitioned log-likelihood $\ln\mathcal{L}(c) = \sum_{u=1}^{c}\ln P(s_{\text{rec}}(u) \mid s_{\text{left}}(u)) + \sum_{u=c+1}^{L}\ln P(s_{\text{rec}}(u) \mid s_{\text{right}}(u))$. Across invariant sites, the likelihood forms a flat plateau. `RhizAeon` reports the exact 95% confidence interval $[c_{\text{left}}, c_{\text{right}}] = \{c : \ln\mathcal{L}(c) \geq \max\ln\mathcal{L} - 1.92\}$, bounding genetic exchange by physical informative mutations. Continuous gap bridging iteratively coalesces adjacent donor fragments separated by non-informative intervals (≤ 150 bp) whenever net flux across the merged span remains positive.

Physical Force Field Deconvolution and Random Matrix Halting

To resolve multi-way mosaics without combinatorial heuristics, `RhizAeon` models trajectory excursions as displacements within a physical force field. The home lineage exerts a Hookean restoring force $\mathbf{f}(u) = -\lambda(\mathbf{z}(u) - \mathbf{z}_{\text{home}}) \in \mathbb{R}^d$. Across candidate breakpoint boundaries, flanking force vectors $\mathbf{f}_{\text{left}}$ and $\mathbf{f}_{\text{right}}$ define step velocity $v_{\text{step}} = \|\mathbf{f}_{\text{right}} - \mathbf{f}_{\text{left}}\|$ and deflection angle $\cos\theta = \langle \mathbf{f}_{\text{left}}, \mathbf{f}_{\text{right}} \rangle / (\|\mathbf{f}_{\text{left}}\| \|\mathbf{f}_{\text{right}}\|)$. Dynamic velocity reversal $R = v_{\text{step}}(1 - \cos\theta)$ captures abrupt momentum changes. Projecting flanking force increments onto unit parental dipole axis $\mathbf{d}_{\text{hat}} = (\mathbf{z}_{\text{donor}} - \mathbf{z}_{\text{home}}) / \|\mathbf{z}_{\text{donor}} - \mathbf{z}_{\text{home}}\|$ yields the scale-invariant test statistic:

$$Z_{\text{flip}} = \frac{|(\mathbf{f}_{\text{right}} - \mathbf{f}_{\text{left}}) \cdot \mathbf{d}_{\text{hat}}|}{\sqrt{\frac{\sigma^2_{\text{left}}}{L_{\text{left}}} + \frac{\sigma^2_{\text{right}}}{L_{\text{right}}}}}. \tag{7}$$

For chromosomes bearing multiple reticulations, spatio-temporal force vectors form matrix $\mathbf{F} \in \mathbb{R}^{N \times L}$. Singular value decomposition $\mathbf{F} = \sum_{m=1}^{r} s_m \mathbf{u}_m \mathbf{v}_m^T$ isolates distinct reticulation modes. Iterative mode peeling subtracts each verified component $s_m \mathbf{u}_m \mathbf{v}_m^T$ from residual forces. Peeling must halt when residual variance reflects neutral coalescent drift. By random matrix theory, singular values of an $N \times L$ noise matrix follow the Marchenko-Pastur distribution (Marchenko and Pastur, 1967). The upper edge of the bulk spectral noise edge is:

$$s_{\text{bulk}} = \frac{L}{\pi} \sigma_{\text{noise}} \left(1 + \sqrt{\frac{P}{L}}\right), \tag{8}$$

where $P = \min(N, d)$ and σ_{noise} is the empirical force increment standard deviation. Peeling terminates when leading residual singular values satisfy $s_m \leq s_{\text{bulk}}$, halting iteration without arbitrary threshold tuning.

Reticulation Graph Adjudication and Ancestral Event Grouping

In reciprocal triplet scans, recombinant isolate C inheriting sequence from donor D into home H frequently triggers reciprocal false alarms (D recombinant between C and H). This reciprocal artifact creates mutual two-cycles in candidate reticulation graphs ($C \leftrightarrow D$). `RhizAeon` resolves orientation by evaluating transition contrast $\Delta_{\text{trans}} = \Delta d_{\text{home}} + \Delta d_{\text{donor}}$, where $\Delta d_{\text{home}} = \max(0, d(C, H \mid \text{tract}) - d(C, H \mid \text{flank}))$. Authentic recombinant children leave their home background within the tract ($\Delta d_{\text{home}} \gg 0$). Stationary clonal parents remain within home background across the alignment ($\Delta d_{\text{home}} \approx 0$). Comparing transition contrast purges reciprocal parent false alarms.

In expanding epidemic outbreaks, multiple descendant isolates inherit identical mosaic tracts from a common recombinant ancestor. Reporting each isolate as an independent recombination event inflates event counts and misrepresents evolutionary history. `RhizAeon` clusters candidate events into shared ancestral recombination events using an interval overlap criterion. Events sharing identical resolved Home and Donor lineages whose breakpoint intervals overlap by $\geq 50\%$ of the shorter tract:

$$\frac{\min(u_2^{(1)}, u_2^{(2)}) - \max(u_1^{(1)}, u_1^{(2)}) + 1}{\min(u_2^{(1)} - u_1^{(1)} + 1,\ u_2^{(2)} - u_1^{(2)} + 1)} \geq 0.50, \tag{9}$$

collapse into a single ancestral event. Grouped events aggregate the complete roster of co-recombinant isolates ($\mathcal{I} = \{s_1, \ldots, s_k\}$). Consensus breakpoints span the outer envelope $[\min_k u_{1,k}, \max_k u_{2,k}]$, while retaining maximal statistical significance across member isolates. Each unique event receives a persistent sequential identifier $(1, 2, \ldots)$, reporting both unique evolutionary events and total recombinant isolates across the alignment inventory.

Coalescent Simulation Suites and Bipartite Hungarian Evaluation

Synthetic alignments were generated under Hudson's neutral coalescent model with recombination (Hudson, 1983; Posada and Crandall, 2001) across 3,200 alignments spanning 16 parameter regimes. Simulations crossed sequence divergence $\theta \in \{2, 10, 50\}$, recombination rate $\rho \in \{0, 4, 16, 64\}$, and rate heterogeneity $\alpha \in \{0.5, \infty\}$. To evaluate operational challenges, four specialized scenario suites were simulated under HKY85 substitution dynamics ($\kappa = 2.5, \pi = [0.3, 0.2, 0.2, 0.3]$) across 2,280 alignments:

1. **Epidemic Clonal Expansion**: Recombinant ancestors sprouting $K \in \{2, 5, 8, 10, 15\}$ descendants with transmission drift ($\delta = 0.002$) alongside 40 parental background isolates ($N = 42$–55, $L = 5,000$ nt; 300 alignments).
2. **Multi-Way Mosaic Swarms**: Dual-cassette transfers, nested Russian doll introgressions, and triple crossover cascades with $K \in \{1, 5, 10\}$ descendants across $d \in \{0.02, 0.05, 0.10\}$ ($N = 46$–55, $L = 6,000$ nt; 540 alignments).
3. **Ghost Donor Introgression**: Pure unsampled donor, surrogate sister clade, and deep reservoir donor regimes across swarm sizes $K \in \{2, 5, 10\}$ ($N = 32$–40, $L = 5,000$ nt; 540 alignments).
4. **Micro-Conversion Resolution**: Conversion tracts spanning $L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt across three divergence tiers ($N = 32$–40, $L = 5,000$ nt; 900 alignments).

In addition, a grand parameter sweep of 150,000 whole genomes was evaluated across continuous distributions: $L \in [1{,}000, 30{,}000]$ nt, $N \in [4, 250]$ taxa, $d \in [0.005, 0.25]$ subs/site, and Gamma rate shape $\alpha \in [0.05, 5.0]$.

Ground truth was evaluated via maximum-weight bijective bipartite matching using the Hungarian algorithm (Kuhn, 1955). For true event $E = (C, H, D, \mathcal{I})$ and inferred event $\hat{E} = (\hat{C}, \hat{H}, \hat{D}, \hat{\mathcal{I}})$, event affinity combines genealogical triad affinity and informative SNP Jaccard similarity:

$$s(E, \hat{E}) = \left[w(C, \hat{C}) \times \frac{w(H, \hat{H}) + w(D, \hat{D})}{2} \right] \times \frac{|\mathcal{S}_{\text{true}} \cap \mathcal{S}_{\text{inferred}}|}{|\mathcal{S}_{\text{true}} \cup \mathcal{S}_{\text{inferred}}|}, \tag{10}$$

where $w(X, Y) = \exp(-d_{\text{tree}}(X, Y)/\sigma_{\text{clade}})$. Across the optimal matching $\mathcal{M}$, reticulation precision and recall are $\text{PPV}_{\mathcal{R}} = \frac{1}{\hat{K}} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$ and $\text{TPR}_{\mathcal{R}} = \frac{1}{K} \sum_{\mathcal{M}} s(E_i, \hat{E}_j)$. The composite Reticulation Fidelity Index is their harmonic mean $\mathcal{R}_{F_1} = 2 \cdot \text{PPV}_{\mathcal{R}} \cdot \text{TPR}_{\mathcal{R}} / (\text{PPV}_{\mathcal{R}} + \text{TPR}_{\mathcal{R}})$. Every unmatched true event penalizes recall; every ungrounded prediction penalizes precision.

Empirical Genomic Datasets and Implementation

Empirical evaluations analyzed three natural evolutionary cohorts:

1. **Sarbecovirus Genomes**: 83 whole genomes of bat, pangolin, and human sarbecoviruses (30,950 nt; Lytras et al. (2022)), including SARS-CoV-2 reference NC_045512.2 and close relatives RaTG13, BANAL-20-52, and RmYN02.
2. **Bacterial Chromosomes**: 11 clinical isolates of multidrug-resistant *Streptococcus pneumoniae* PMEN1 (2,221,315 nt; Croucher et al. (2011)) spanning global penicillin-resistant outbreaks.
3. **Primate Immunogenomics**: 379 classical primate MHC Class I alleles (*HLA-A*, *HLA-B*, *HLA-C*, and non-human hominoid orthologs; 1,130 nt; Maccari et al. (2017)).

`RhizAeon` is implemented in native Rust with Rayon multicore parallelization and PyO3 Python bindings. The algorithmic core compiles to WebAssembly (WASM). Surveillance pipelines execute client-side within standard web browsers without remote server transmission. All computation executes locally on client hardware without transmitting sequence data across external networks. Interactive dashboards render synchronized signal inspectors, 2D metric subspace projections, and midpoint-rooted phylogenetic trees within browser canvas. Source code, documentation, and benchmark workflows are freely available under the MIT license at `https://github.com/veg/rhizaeon-code`.

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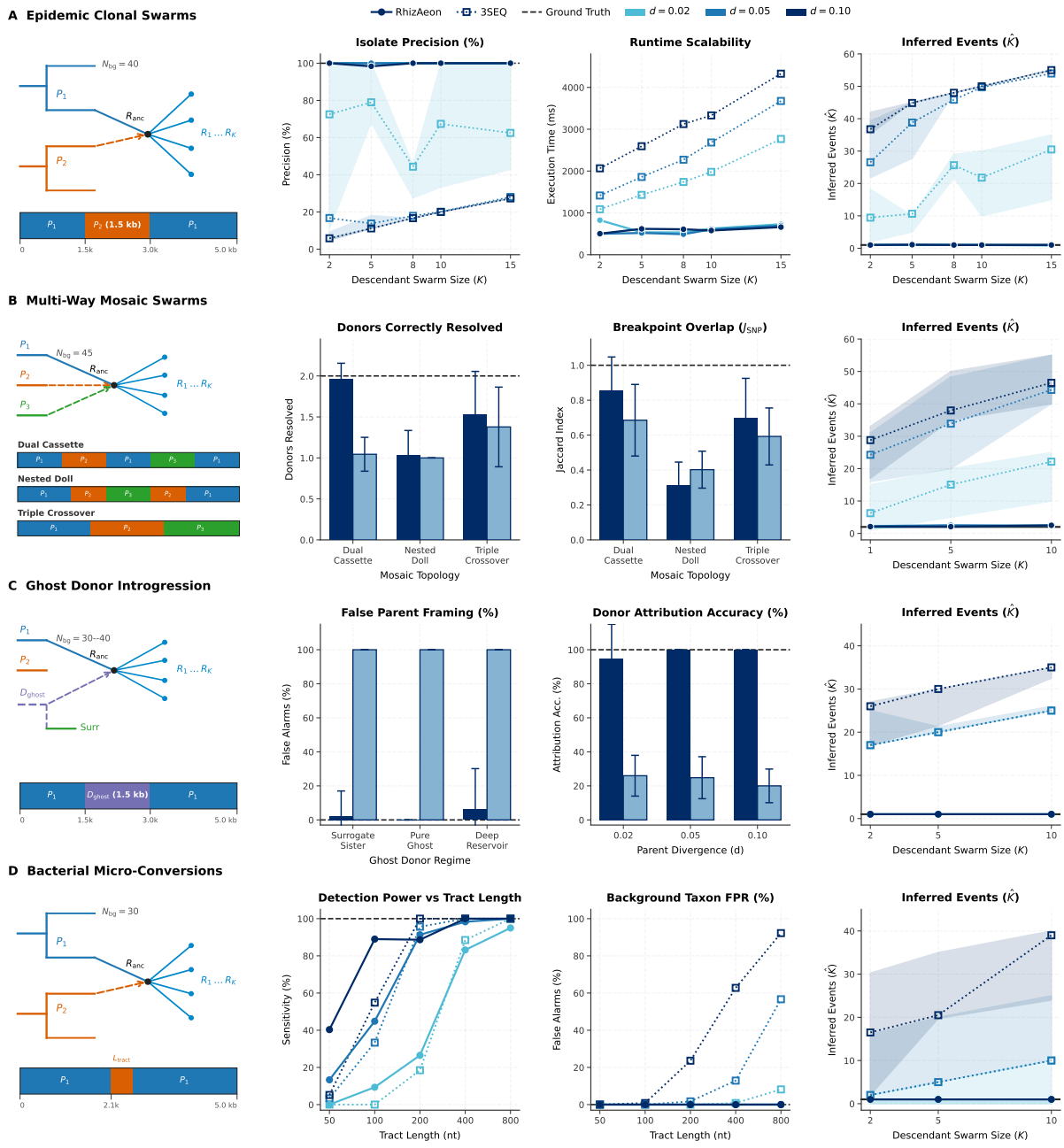


Figure 2 Operational performance across specialized genomic and epidemiological regimes. Each row depicts the simulation topology (Col. 1), primary diagnostic metric (Col. 2), boundary precision (Col. 3), and inferred cardinality ($\hat{K}$) across descendant swarm sizes (K, Col. 4). **(A)** Epidemic Clonal Expansion: An ancestral reticulation ($P_1, P_2 \to R_{\text{anc}}$) generates a 1.5 kb cassette in $K \in \{2\text{–}15\}$ isolates. While 3SEQ inflates events ($\hat{K} = 9.5\text{–}55.0$) and scales linearly, `RhizAeon` collapses swarms into a single event ($\hat{K} = 1.00$) with 99.9% precision in flat runtime ($491\text{–}826$ ms). **(B)** Multi-Way Mosaic Swarms: Dual Cassette, Nested Doll, and Triple Crossover architectures ($K_{\text{donor}} = 2.0$). `RhizAeon` resolves both parental donors ($1.97/2.00$) with high boundary overlap ($J = 0.86\text{–}0.98$), whereas closed-world triplet scanning misses secondary donors and inflates events up to $\hat{K} = 46.5$. **(C)** Ghost Donor Introgression: Incomplete taxon sampling across Surrogate Sister, Pure Ghost, and Deep Reservoir regimes. 3SEQ misidentifies background parent P_2 as donor in 100% of trials. `RhizAeon` maintains $\leq 6\%$ false framing and attributes 98.3% of surrogate donors correctly. **(D)** Bacterial Micro-Conversions: Short replacement tracts ($L_{\text{tract}} \in [50, 800]$ nt). Continuous dyadic wavelets achieve 40.3% power at 50 nt ($d = 0.10$, vs. 3SEQ 5.2%) and 89.0% at 100 nt, with 0.00% background false alarms.

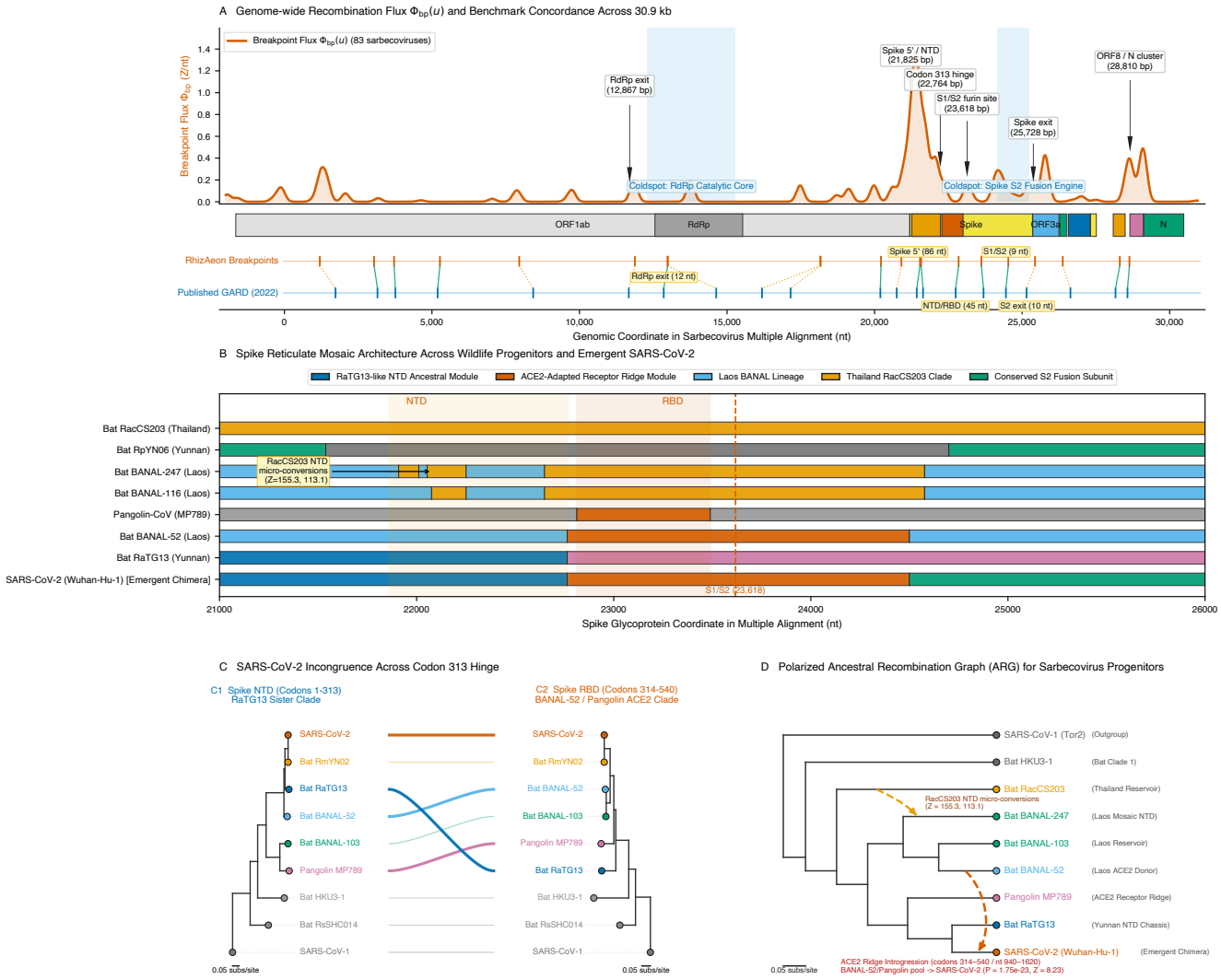


Figure 3 Sarbecovirus whole-genome recombination landscapes, Spike mosaicism, and ancestral recombination graphs. A. Genome-wide breakpoint flux $\Phi_{\mathrm{bp}}(u)$ across 83 sarbecoviruses (30.9 kb) aligned to gene annotations. Recombination hotspots concentrate at the Spike $5'$ NTD junction (21,825 bp), codon 313 hinge (22,764 bp), S1/S2 furin site (23,618 bp), and ORF8/N cluster (28,810 bp). Coldspots protect the RdRp catalytic core (nt 13,400–16,200) and Spike S2 fusion machinery. Beneath the genes, `RhizAeon` changepoints (orange) match 21 published GARD breakpoints (blue) with 99.0 nt median deviation. **B.** Reticulate Spike glycoprotein mosaicism (nt 21,000–26,000). Wuhan-Hu-1 is deconvolved as a chimera: a bat RaTG13-like NTD (blue, 94.4% identity), an ACE2-adapted receptor-binding ridge shared with Laotian bat BANAL-52 and pangolin MP789 (vermillion), and a conserved S2 core (green). **C.** Segmental incongruence across codon 313 in a two-partition tanglegram. Wuhan-Hu-1 clusters sister to RaTG13 in the NTD (C1), whereas in the RBD (C2), BANAL-52 and pangolin MP789 shift sister to Wuhan-Hu-1. **D.** Polarized Ancestral Recombination Graph (ARG). Dashed arcs trace introgression of the human-compatible ACE2 ridge into ancestral SARS-CoV-2 (SSCR 94.2% vs. 62.1%, $P = 1.75 \times 10^{-23}$) and NTD micro-conversions from Thai RacCS203 into BANAL-247. The tripartite ARG reduces residual strain by 74.0% ($\Delta\mathrm{BIC} = 0.0$).

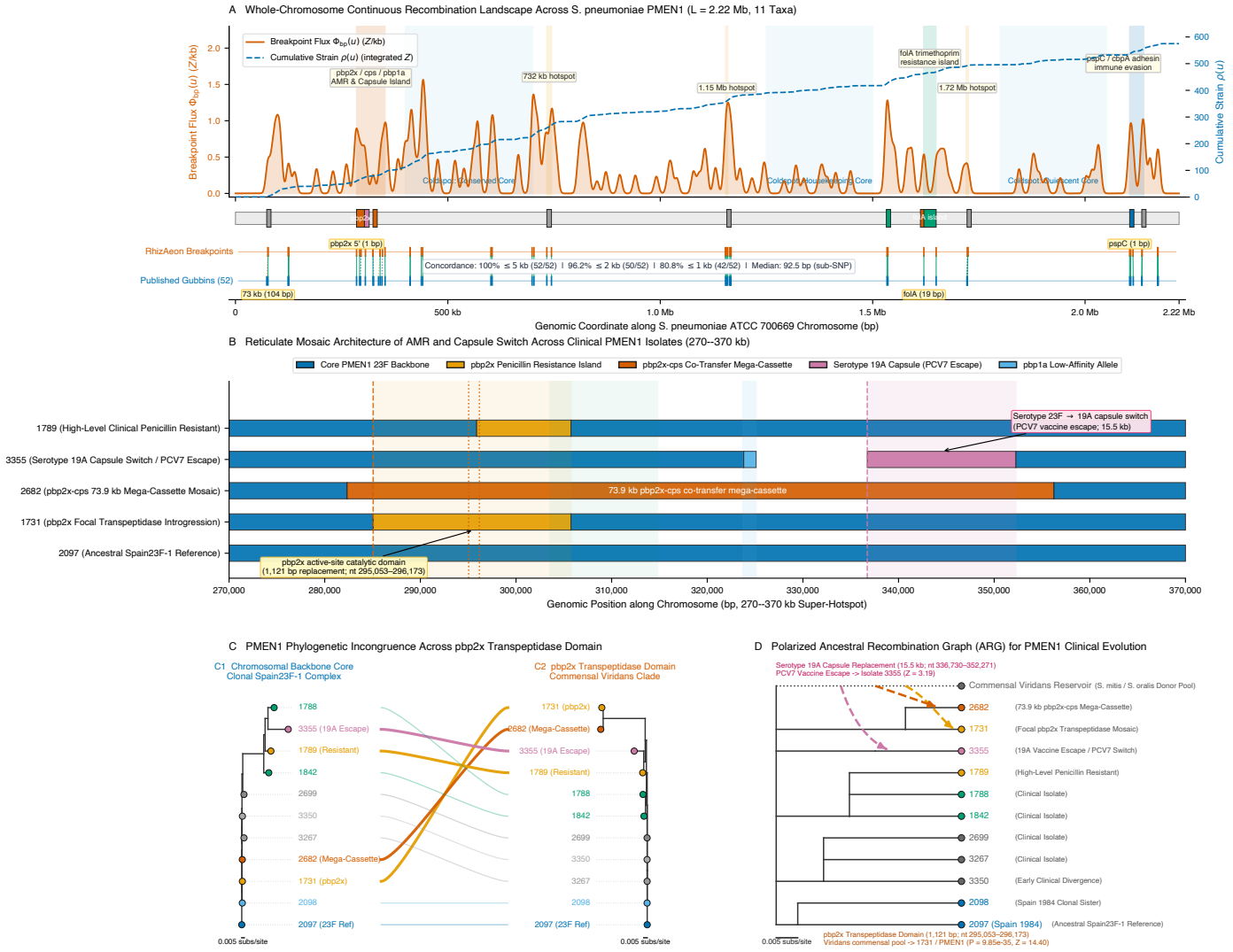


Figure 4 Whole-chromosome recombination deconvolution, antimicrobial resistance mosaicism, and ancestral recombination graphs in Streptococcus pneumoniae PMEN1. A. Chromosomal breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across 11 global clinical isolates (2.22 Mb, Spain23F-1 / PMEN1). Six hotspots emerge: $pbp2x/cps$ (285–352 kb), 732 kb, 1.15 Mb, *folA* (1,620–1,649 kb), 1.72 Mb, and *pspC/cbpA* (2,105–2,138 kb). Tickmarks benchmark `RhizAeon` changepoints (orange) against 52 published Gubbins boundaries (blue; 96.2% concordance ≤ 2 kb, median deviation 92.5 nt). **B.** AMR and capsular switch mosaicism (nt 270,000–370,000). Ancestral isolate 2097 carries the core backbone (dark blue); 1731 displays the 1,121 bp active-site *pbp2x* transpeptidase replacement ($P = 9.85 \times 10^{-35}$); 2682 captures a 73.9 kb *pbp2x-cps* co-transfer mega-cassette; and 3355 reveals serotype 23F $\rightarrow$ 19A capsule replacement enabling PCV7 vaccine escape. **C.** Segmental incongruence across the *pbp2x* catalytic domain. In the core chromosome, all isolates form a monophyletic complex; in *pbp2x*, isolates 1731 and 2682 diverge into commensal viridans streptococci (*S. mitis* / *S. oralis*, $\Delta d > 6\%$). **D.** Polarized ARG for PMEN1 clinical adaptation. Arcs reconcile *pbp2x* active-site import, 73.9 kb co-transfer, and 19A capsular replacement onto the Spain23F-1 backbone.

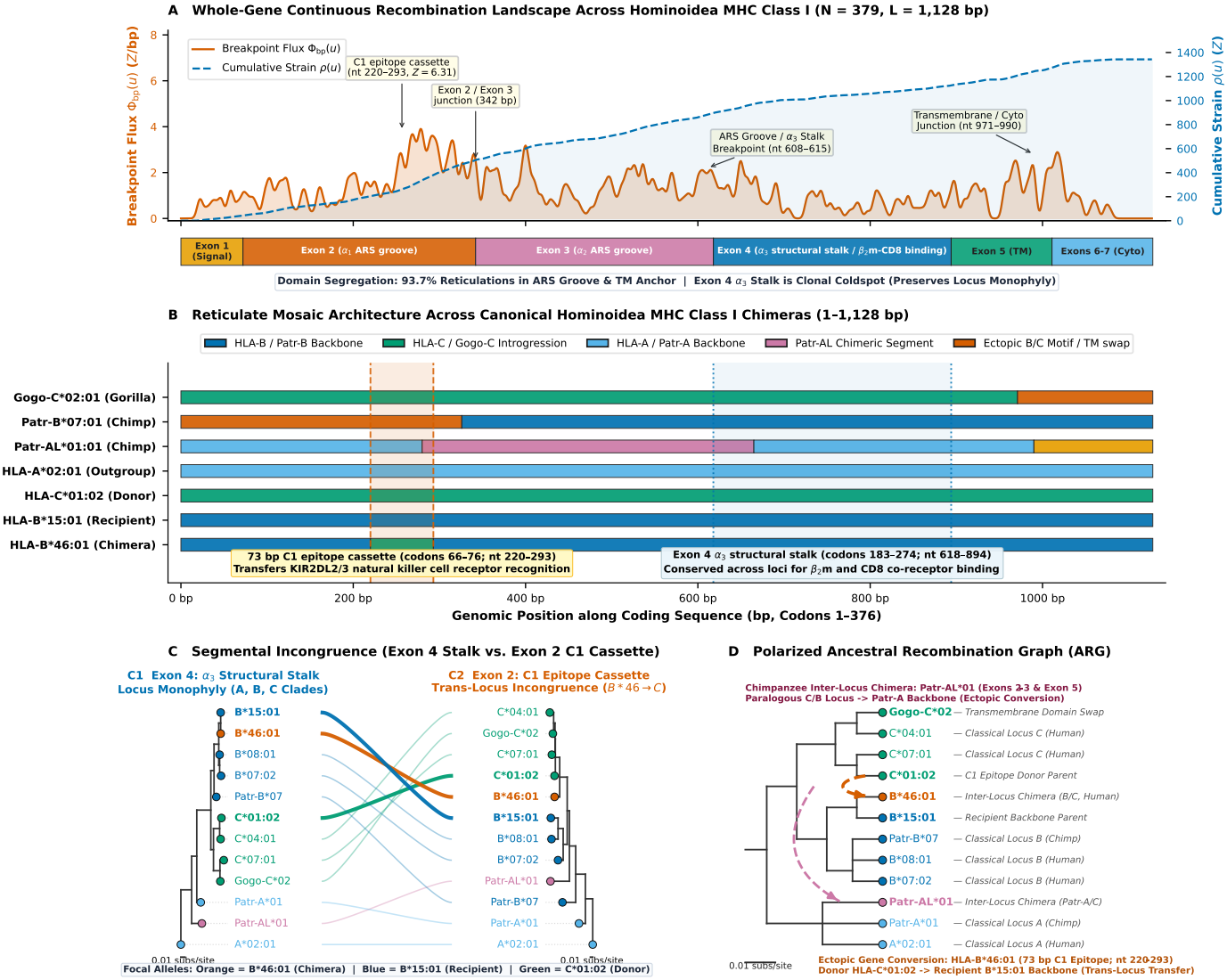


Figure 5 Whole-gene recombination deconvolution, ectopic micro-conversion, and ancestral recombination graphs in Hominoidea MHC Class I. A. Breakpoint flux $\Phi_{\mathrm{bp}}(u)$ (vermillion) and cumulative strain $\rho(u)$ (dashed blue) across $N = 379$ primate alleles (1,128 bp, Exons 1–7). Reticulations concentrate in the antigen recognition site (ARS; Exons 2–3, nt 72–618) and transmembrane anchor (Exon 5), while Exon 4 (α_3 stalk) remains a conserved coldspot (93.7% of strain in ARS and TM). The unsupervised boundary inflection at nt 608–615 matches the 618 bp Exon 3/4 junction. **B.** Reticulate mosaic architecture in primate MHC Class I chimeras. *HLA-B*46:01* is deconvolved as an inter-locus chimera carrying an exact 73 bp micro-conversion (nt 220–293, codons 66–76; green) from *HLA-C*01:02* within an *HLA-B*15:01* backbone (dark blue), transferring the C1 functional epitope that binds NK cell receptors KIR2DL2/3. **C.** Segmental incongruence across the α_3 boundary. In Exon 4, *HLA-A*, *B*, and *C* form monophyletic clades (*B*46:01* sister to *B*15:01*). In the Exon 2 C1 cassette, *B*46:01* translocates into the *HLA-C* clade sister to *C*01:02*. **D.** Polarized ARG for primate MHC. Reticulation arcs anchor the 73 bp C1 micro-conversion and ancient chimpanzee *Patr-AL*01* transfers onto the tandem gene duplication backbone.

Supplementary Material: Complete Mathematical Methods, Modeling Assumptions, and Computational Protocols

Foundational Modeling Assumptions and Physical Boundary Conditions

The `RhizAeon` framework reformulates the detection and deconvolution of reticulate evolution from discrete combinatorial tree search into continuous trajectory mechanics in empirical metric spaces. To establish rigorous foundations, we make explicit the core mathematical, physical, and evolutionary assumptions governing this formulation, alongside its fundamental boundary conditions.

Assumption 1 (Metric Embeddability of Evolutionary Divergence).

We assume that pairwise genetic distances computed across nucleotide or amino acid alignments can be embedded into a low-dimensional Euclidean metric space $(\mathbb{R}^k, \|\cdot\|_2)$ with minimal topological distortion. For any two sequences i and j, their global evolutionary divergence $d_{ij} \in [0,1]$ represents an additive metric over tree branches under tree-like evolution, satisfying the triangle inequality $d_{ij} \leq d_{ik} + d_{kj}$ up to finite-sites homoplasy. Through classical multidimensional scaling (MDS), the centering transformation $\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}^{\odot 2})\mathbf{H}$ yields a positive semi-definite inner product matrix on the subspace of dominant eigenvalues. Gower's out-of-sample projection guarantees that any local sub-interval distance vector $\mathbf{d}_i(u)$ maps bijectively into this canonical coordinate frame without recalculating eigenvectors or requiring Procrustean superimposition.

Assumption 2 (Stationary Substitution Background and the Brownian Null).

We assume that under purely clonal, non-recombinant evolution, nucleotide substitutions accumulate along lineages according to a continuous-time Markov substitution process with stationary distribution $\boldsymbol{\pi}$ and rate matrix $\mathbf{Q}$. Scanning along an alignment from $5'$ to $3'$, the cumulative difference between any two clonal lineages accumulates as a sum of independent, identically distributed Bernoulli substitution trials. By Donsker's invariance principle, as alignment length $L \to \infty$, the standardized cumulative deviation curve:

$$B(u) = \frac{Y(u) - \frac{u}{L}Y(L)}{\sigma\sqrt{L}} \tag{S1}$$

converges weakly to a standard Brownian bridge on $u \in [0,1]$, where $\mathbb{E}[B(u)] = 0$ and $\mathrm{Cov}(B(s), B(t)) = \min(s,t) - st$. Under the null hypothesis of clonal evolution (H_0), fluctuations in sequence affinity represent balanced thermal noise. Recombination is formally defined as an exogenous physical force that breaks this thermal equilibrium, inducing a sustained, non-zero drift velocity $\frac{d}{du}Y(u) \neq 0$ that exceeds Brownian boundary-crossing thresholds.

Assumption 3 (The Unpolarized Root Token as an Autapomorphic Sink).

Mutational rate heterogeneity (heterotachy) and directional hypermutation bursts (such as APOBEC, ADAR, or error-prone replication) frequently generate localized clusters of mutations along a single lineage. Heuristic triplet methods mistake these mutational clusters for horizontal donor acquisition. `RhizAeon` eliminates this vulnerability by reserving coordinate index 0 in the cross-attention simplex for an unpolarized ancestral root token $\mathbf{x}_{\text{root}}$, anchored at the metric origin $\mathbf{m}_{\text{root}} = \mathbf{0}$. In the cross-attention softmax:

$$A_u(i \to \text{root}) + \sum_{j=1}^{N} A_u(i \to j) = 1.0. \tag{S2}$$

When lineage i acquires private substitutions that do not point toward any sampled contemporary taxon, the similarity logits to all contemporary leaves decrease uniformly, naturally routing attention into the root sink at index 0. Consequently, autapomorphic bursts reduce contemporary affinity without surging candidate donor attention, providing intrinsic mathematical immunity to false positive alarms under rate heterogeneity.

Assumption 4 (Spectral Landmark Representation on Simplex Δ^K).

To scale linearly across vast genomic cohorts ($N >$ 1,000 genomes), `RhizAeon` assumes that the evolutionary variance of a population can be faithfully captured by K spectral landmark centroids ($K \ll N$) on the probability simplex Δ^K. Landmarks are selected via k-medoids clustering on global distance matrix $\mathbf{D}_{\text{glob}}$, ensuring uniform coverage of deep clades. Every contemporary genome i is tracked as a continuous barycentric coordinate vector $\boldsymbol{\alpha}_i(u) \in \Delta^K$. This reduces computational complexity from $\mathcal{O}(N^2L)$ or $\mathcal{O}(N^3L)$ down to $\mathcal{O}(NKL)$ while preserving sensitivity to cross-clade transfers.

Physical Limits and Resolution Boundaries.

The resolution of trajectory deconvolution is governed by fundamental physical boundaries:

1. *Mutational Density Floor*: Recombination can only be inferred if the introgressed tract contains distinguishing polymorphic mutations between parental lineages ($n_{\text{mut}} = L_{\text{tract}} \cdot d_{\text{parent}} \geq 1$). In low-divergence regimes ($d_{\text{parent}} < 0.005$), statistical power is fundamentally bounded by mutation availability rather than algorithmic sensitivity.
2. *Tract Length Resolution*: The dyadic scale-space matched filter operates across dyadic scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt. Replacement tracts shorter than 15 nt (single-codon swaps) cannot be reliably distinguished from multi-nucleotide mutations without prior structural constraints.

3. *Extinction / Ghost Limits*: When an introgressing donor has no sampled relatives within phylogenetic distance $d_{\max}$, the tract is classified as an orphan ghost introgression. Directional consistency constraints prevent false attribution to contemporary outgroups, but donor clade identification remains impossible without surrogate sampling.

Complete Algorithmic Pipeline and Mathematical Formulations

Lossless SNP Compression and Prefix Distance Tensor

Given a multiple sequence alignment $\mathbf{M} \in \Sigma^{N \times L}$ over alphabet $\Sigma = \{A, C, G, T, -\}$, columns exhibiting zero polymorphic variation across all N taxa carry zero phylogenetic information. `RhizAeon` indexes out all monomorphic positions, compressing alignment $\mathbf{M}$ into a segregating variant matrix $\mathbf{S} \in \Sigma^{N \times L_{\text{SNP}}}$ with bijective coordinate mapping function $\psi : \{0, \ldots, L_{\text{SNP}} - 1\} \to \{0, \ldots, L - 1\}$. For whole-genome bacterial alignments where $L_{\text{SNP}} \ll L$ (1,465-fold compression in *S. pneumoniae*), this allows all trajectory computations to execute in memory-resident cache.

Across the compressed alignment, we compute the cumulative prefix difference tensor $\mathbf{\Phi} \in \mathbb{R}^{N \times N \times (L_{\text{SNP}}+1)}$:

$$\Phi_{ij}(u) = \sum_{c=0}^{u-1} w_c \cdot \mathbb{I}(S_{ic} \neq S_{jc}), \tag{S3}$$

where w_c denotes site-specific weights (e.g., transition/transversion corrections or codon degeneracy). For any genomic sub-interval $[u_1, u_2)$, the exact pairwise evolutionary distance between sequence i and sequence j is computed in $\mathcal{O}(1)$ time:

$$D_{ij}(u_1, u_2) = \frac{\Phi_{ij}(u_2) - \Phi_{ij}(u_1)}{u_2 - u_1}. \tag{S4}$$

Global Divergence Matrix and Classical MDS Basis

The whole-sequence evolutionary divergence matrix $\mathbf{D}_{\text{glob}} \in \mathbb{R}^{N \times N}$ is evaluated across the full locus: $D_{\text{glob},ij} = \Phi_{ij}(L_{\text{SNP}})/L_{\text{SNP}}$. We construct the centering projection matrix $\mathbf{H} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^T$ and form the Gower double-centered inner product matrix:

$$\mathbf{B} = -\frac{1}{2}\mathbf{H}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{H}. \tag{S5}$$

Spectral decomposition yields $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, where eigenvalues are sorted in descending order $\lambda_1 \geq \cdots \geq \lambda_N \geq 0$. We retain the top $k = \min(3, N-1)$ positive eigenvalues $\mathbf{\Lambda}_k = \text{diag}(\lambda_1, \ldots, \lambda_k)$ and corresponding orthonormal eigenvectors $\mathbf{V}_k \in \mathbb{R}^{N \times k}$. This canonical basis is computed once and held fixed for the entire alignment, guaranteeing zero Procrustes rotational drift.

Exact Gower Out-of-Sample Continuous Trajectory Projection

At any genomic coordinate $u \in [0, L_{\text{SNP}})$, each sequence i possesses an instantaneous local distance vector $\mathbf{d}_i(u) \in \mathbb{R}^N$ to all reference taxa, evaluated over a symmetric Gaussian kernel window:

$$d_i(j; u)^2 = \sum_{c=0}^{L_{\text{SNP}}-1} K_\sigma(c - u) \cdot \mathbb{I}(S_{ic} \neq S_{jc}) \Big/ \sum_{c=0}^{L_{\text{SNP}}-1} K_\sigma(c - u). \tag{S6}$$

The exact Gower out-of-sample projection maps this local distance vector directly into canonical Euclidean space $\mathbb{R}^k$:

$$\mathbf{z}_i(u) = \frac{1}{2}\mathbf{\Lambda}_k^{-1/2}\mathbf{V}_k^T \left(\bar{\mathbf{d}}^2 - \mathbf{d}_i(u)^2\right), \tag{S7}$$

where $\bar{\mathbf{d}}^2 = \frac{1}{N}(\mathbf{D}_{\text{glob}}^{\odot 2})\mathbf{1}_N$ is the precomputed vector of average squared global distances. Evaluating $\mathbf{z}_i(u)$ across all u produces the continuous trajectory tensor $\mathbf{Z} \in \mathbb{R}^{N \times L_{\text{SNP}} \times k}$.

Continuous Phylogenetic Cross-Attention with Tree-RoPE and PhyloBias

To track affinity dynamics against reference landmarks without heuristic window slicing, `RhizAeon` employs continuous cross-attention. For each position u, sequence tokens are embedded into query ($\mathbf{q}_i(u)$), key ($\mathbf{k}_j(u)$), and value ($\mathbf{v}_j(u)$) representations in $\mathbb{R}^d$. Phylogenetic distance constraints are injected through two mechanisms:

1. *Tree-RoPE (Tree Rotary Position Embedding)*: Metric MDS coordinates $\mathbf{m}_i \in \mathbb{R}^4$ modulate query and key vectors via multidimensional Givens rotations:

$$\tilde{\mathbf{q}}_{i,2m:2m+1}(u) = \mathbf{R}(\theta_{i,m})\mathbf{q}_{i,2m:2m+1}(u), \qquad \theta_{i,m} = \sum_{c=1}^{4} m_{ic}\Theta_{mc}. \tag{S8}$$

2. *PhyloBias*: A log-space continuous-time Markov substitution prior penalizes attention between divergent lineages:

$$\text{PhyloBias}_{ij} = \ln\left[\epsilon_0 + (1-\epsilon_0)\exp(-\lambda_h D_{\text{glob},ij})\right], \tag{S9}$$

where $\epsilon_0 = 0.05$ represents the neutral substitution floor.

The resulting attention tensor $\mathbf{A} \in \mathbb{R}^{L\times N\times(N+1)}$ evaluates attention from each query sequence i to all reference taxa $j \in \{1,\ldots,N\}$ and the root sink at index 0:

$$A_u(i\to j) = \frac{\exp\left(\frac{\tilde{\mathbf{q}}_i(u)^T\tilde{\mathbf{k}}_j(u)}{\sqrt{d}} + \text{PhyloBias}_{ij}\right)}{\sum_{l=0}^{N}\exp\left(\frac{\tilde{\mathbf{q}}_i(u)^T\tilde{\mathbf{k}}_l(u)}{\sqrt{d}} + \text{PhyloBias}_{il}\right)}. \tag{S10}$$

Cumulative Attention Curves and Differential Drift

For each sequence i and reference landmark bundle m, instantaneous attention is accumulated along the chromosome in $\mathcal{O}(L)$ time:

$$C_{im}(u) = \sum_{c=0}^{u}\alpha_{im}(c), \qquad \text{where} \quad \alpha_{im}(c) = \sum_{j\in\mathcal{B}_m} A_c(i\to j). \tag{S11}$$

The home clade $\mathcal{B}_{\text{home}}(i)$ is identified as the majority recipient across the full locus: $\mathcal{B}_{\text{home}}(i) = \arg\max_m C_{im}(L)$. The differential attention curve against candidate donor m is formulated as:

$$Y_{im}(u) = C_{im}(u) - C_{i,\text{home}}(u). \tag{S12}$$

Under clonal evolution, sequence i maintains stable affinity to its home clade, resulting in negative or stationary drift $Y_{im}(u) \le 0$. A recombination event from donor m produces a positive velocity excursion where $\frac{d}{du}Y_{im}(u) > 0$.

Multi-Scale Dyadic Scale-Space Matched Filter and Running Drawdown

To detect both focal micro-conversions and macroscopic recombinant cassettes without parameter tuning, differential drift increments $\Delta Y_{im}(u) = Y_{im}(u) - Y_{im}(u-1)$ are convolved with zero-mean bipolar matched filter wavelets across dyadic scales $\tau \in \{15, 25, 50, 100, 200, 500\}$ nt:

$$W_\tau(u) = \sum_{s=-\tau}^{\tau}\Delta Y_{im}(u+s)\cdot\psi_\tau(s), \qquad \psi_\tau(s) = \frac{s}{\tau}\exp\left(-\frac{s^2}{2\tau^2}\right). \tag{S13}$$

Concurrently, macroscopic cassettes are identified via Taylor running drawdown:

$$M(u) = \max_{0\le s\le u} Y_{im}(s), \qquad \text{DD}(u) = M(u) - Y_{im}(u). \tag{S14}$$

A candidate cassette is detected when the excursion drawdown exceeds the extreme-value threshold $Z_{\text{phys}} \ge Z_{\text{crit}}$, where $Z_{\text{crit}} = \sqrt{-0.5\ln(\alpha/2M)}$ by Kolmogorov supremum boundary crossing.

Profile Maximum-Likelihood Breakpoint Polisher

Candidate changepoint intervals $[u_1-\delta, u_1+\delta]$ and $[u_2-\delta, u_2+\delta]$ localized by the matched filter are polished to single-nucleotide resolution using profile maximum likelihood. For each candidate coordinate b, the alignment is cleaved into left ($[0,b)$) and right ($[b,L)$) partitions. The profile likelihood ratio:

$$\Delta\ln L(b) = \ln P(\mathbf{S}_{[0,b)} \mid T_{\text{home}}) + \ln P(\mathbf{S}_{[b,L)} \mid T_{\text{donor}}) - \ln P(\mathbf{S}_{[0,L)} \mid T_{\text{home}}) \tag{S15}$$

is evaluated under HKY85 substitution. The exact breakpoint $\hat{b}$ is localized at the maximum likelihood peak: $\hat{b} = \arg\max_b \Delta\ln L(b)$. Significance is certified against the Dunn-Šidák corrected family-wise threshold $\alpha_{\text{FWER}} = 1-(1-\alpha)^{1/M}$.

Ancestral Recombination Graph (ARG) Reconstruction and Validation

Ancestral Event Consolidation

Raw recombinant tracts detected across contemporary genomes $\mathcal{E}_{\text{raw}} = \{e_1,\ldots,e_M\}$ are clustered to distinguish ancestral gene transfers from recurrent hotspot activity. Two events e_a and e_b are grouped into ancestral cluster $\mathcal{C}$ if they share compatible donor clades and satisfy spatial boundary tolerance:

$$J(I_a, I_b) = \frac{|I_a \cap I_b|}{|I_a \cup I_b|} \ge \tau_{\text{jaccard}} \quad \text{or} \quad \max(|u_{1,a}-u_{1,b}|, |u_{2,a}-u_{2,b}|) \le \Delta_{\text{boundary}}, \tag{S16}$$

where default thresholds are $\tau_{\text{jaccard}} = 0.60$ and $\Delta_{\text{boundary}} = 100$ nt. The ancestral event coordinates are defined by median breakpoints $\bar{u}_1 = \text{median}(\{u_{1,i}\})$ and $\bar{u}_2 = \text{median}(\{u_{2,i}\})$. When $|\mathcal{C}| > 1$, the transfer is parameterized as a shared ancestral reticulation node ancestral to all member isolates; when $|\mathcal{C}| = 1$, it is parameterized as a private terminal event.

Structural Model Classes

A candidate evolutionary history H is parameterized by an ordered partitioning of the alignment $[0, L)$ into K intervals $\{[s_k, e_k)\}_{k=1}^{K}$ and an associated graph topology:

1. *Clonal Null* (H_0): A single unpartitioned tree across the full locus ($K = 1$), requiring $2N - 3$ branch lengths and 4 substitution parameters ($k = 2N + 1$).
2. *Unpolarized Partitioned Trees* ($H_{\text{partition}}$): Independent unconstrained trees across all K partitions, requiring $k = K(2N - 3) + 4K$ free parameters.
3. *Directed Polarized ARG* (H_{ARG}): A single ancestral backbone tree where reticulations are represented as directed transfer edges that re-attach the recombinant clade inside the transfer interval $[u_1, u_2]$. Each reticulation edge adds only 2 free parameters (source node and target branch attachment) rather than $2N - 3$ unconstrained branch lengths, yielding $k = K_{\text{backbone}}(2N - 3) + 2|\mathcal{E}_{\text{retic}}| + 4$.
4. *Modular Functional Architectures*: Partition models defined by structural protein domains (e.g., Spike NTD, ACE2 receptor-binding domain, and S2 fusion core).

Multidimensional Statistical Validation Engine

Competing evolutionary histories are evaluated across four orthogonal statistical criteria:

1. *Phylogenetic Profile Likelihood and Information Criteria*: Across each partition $k \in \{1, \ldots, K\}$, log-likelihood $\ln L_k = \ln P(\mathbf{Y}_k \mid T_k, \mathbf{Q}, \boldsymbol{\pi})$ is computed via vectorized Felsenstein pruning. Total log-likelihood $\ln L = \sum_k \ln L_k$ is penalized under finite-sample corrected AICc and BIC:
$$\text{AICc} = 2k - 2\ln L + \frac{2k(k+1)}{L - k - 1}, \qquad \text{BIC} = k \ln L - 2 \ln L. \tag{S17}$$
2. *Residual Non-Tree Manifold Strain*: The expected distance matrix is the length-weighted mixture of local partition patristic distance matrices:
$$\mathbf{D}_{\text{model}} = \sum_{k=1}^{K} \frac{e_k - s_k}{L} \mathbf{D}^{(k)}_{\text{patristic}}. \tag{S18}$$
The residual non-tree distortion matrix $\mathbf{R} = \mathbf{D}_{\text{obs}} - \mathbf{D}_{\text{model}}$ yields the residual strain ratio:
$$\mathcal{S}_{\text{strain}} = \frac{\|\mathbf{R}\|_F}{\|\mathbf{D}_{\text{obs}}\|_F}. \tag{S19}$$
The percentage of non-tree strain explained by history H is $(1 - \mathcal{S}_{\text{strain}}) \times 100\%$.
3. *Spectral SNR over Random Matrix Bulk Edge*: Matrix $\mathbf{R}$ is centered via double-centering $\mathbf{B}_R = -\frac{1}{2}\mathbf{H}(\mathbf{R}^{\odot 2})\mathbf{H}$. The leading eigenvalue $s_1 = \lambda_{\max}(\mathbf{B}_R)$ is compared against the theoretical Marchenko-Pastur bulk noise edge:
$$s_{\text{bulk}} = \sigma^2_{\text{res}} \left(1 + \sqrt{\frac{N}{L}}\right)^2, \tag{S20}$$
where σ_{res} is the standard deviation of off-diagonal elements in $\mathbf{R}$. An empirical SNR $= s_1/s_{\text{bulk}} \leq 1.0$ certifies that residual strain has collapsed into the bulk noise distribution.
4. *Synapomorphic Site Concordance Ratio (SSCR) and Polarity*: At each polymorphic site u where donor D and home H carry distinct nucleotides ($D_u \neq H_u$), recombinant allele R_u is scored as donor-concordant ($R_u = D_u$) or home-concordant ($R_u = H_u$):
$$\text{SSCR}_{\text{in}} = \frac{n_{\text{in}}(R = D)}{n_{\text{in}}(R = D) + n_{\text{in}}(R = H)}, \qquad \text{SSCR}_{\text{flank}} = \frac{n_{\text{flank}}(R = D)}{n_{\text{flank}}(R = D) + n_{\text{flank}}(R = H)}. \tag{S21}$$
Polarization contrast is $\Delta\text{SSCR} = \text{SSCR}_{\text{in}} - \text{SSCR}_{\text{flank}}$, with significance evaluated via a one-sided Fisher's exact test on $[[n_{\text{in}}(D), n_{\text{in}}(H)], [n_{\text{flank}}(D), n_{\text{flank}}(H)]]$.

Coalescent ARG Simulation Protocol

Genealogies and ancestral recombination graphs (ARGs) were simulated using Hudson's continuous-time coalescent framework with homologous recombination (Hudson, 1983; Posada and Crandall, 2001). For a sample of $n = 10$ contemporary lineages across an alignment of length $L = 1{,}000$ nucleotides, the coalescent process tracks active ancestral segments backward in time from $t = 0$. At any time t, let k denote the number of active ancestral chromosomes, and let g_i denote the genomic span between the leftmost and rightmost ancestral segments carried by lineage i, with total ancestral span $G = \sum_{i=1}^{k} g_i$.

The waiting time Δt to the next evolutionary event is exponentially distributed with total event rate $\lambda_{\text{tot}} = \lambda_{\text{coal}} + \lambda_{\text{rec}}$, where:
$$\lambda_{\text{coal}} = \frac{k(k-1)}{2}, \qquad \lambda_{\text{rec}} = \frac{\rho \cdot G}{2L}. \tag{S22}$$
The population recombination parameter is defined as $\rho = 4N_e r$, where N_e is the effective diploid population size and r is the per-generation recombination probability across the L nucleotide locus. With probability $\lambda_{\text{coal}}/\lambda_{\text{tot}}$, a coalescence event merges two randomly chosen lineages

into an ancestral node. With probability $\lambda_{\text{rec}}/\lambda_{\text{tot}}$, a recombination event selects an eligible lineage with probability proportional to its span g_i and cleaves it at a uniform breakpoint coordinate $b \in [u_{\text{start}}, u_{\text{end}}]$, partitioning ancestral material into left and right lineages. Coalescent tracking halts when all L nucleotide sites reach their respective MRCA, yielding an ordered sequence of P genealogical partitions, each defined by an interval $[s_p, e_p)$ and a local gene tree T_p.

Nucleotide sequences were evolved along the branch lengths of each local gene tree T_p under the continuous-time Markov HKY85 substitution model (Hasegawa et al., 1985). The model parameterizes unequal base frequencies $\boldsymbol{\pi} = [0.40, 0.20, 0.30, 0.10]$ and transition/transversion bias $\kappa = 2.0$. The infinitesimal generator matrix $\mathbf{Q} = (Q_{ij})$ is defined by:

$$Q_{ij} = \begin{cases} \kappa\mu\pi_j & \text{for transitions } (A \leftrightarrow G, C \leftrightarrow T), \\ \mu\pi_j & \text{for transversions,} \\ -\sum_{k \neq i} Q_{ik} & \text{for } i = j, \end{cases} \tag{S23}$$

normalized such that average substitution rate $-\sum_i \pi_i Q_{ii} = 1.0$. Branch lengths t_b generated in coalescent units were scaled by the population mutation parameter $\theta = 4N_e\mu \in \{10, 50, 100, 200\}$ so that the effective branch length is $b = t_b \cdot \theta/(2L)$. Site-rate heterogeneity was introduced by drawing independent site mutation rate multipliers $r_u \sim \text{Gamma}(\alpha, 1/\alpha)$ across $u \in [0, L)$, such that $\mathbb{E}[r_u] = 1.0$ and $\text{Var}(r_u) = 1/\alpha$, with shape parameter varying across $\alpha \in \{2.0, 0.5, 0.05\}$.

Specialized Epidemiological and Genomic Simulation Suites

Four specialized simulation generators probed operational regimes outside the neutral coalescent (Figure 2):

- **Epidemic Clonal Expansion**: Trees were constructed by generating a bifurcating clonal backbone of 40 background parent taxa across three parental divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$). An ancestral recombination event was introduced by transferring a 1,501 nt cassette (coordinates 1,500–3,000) from Parent 2 to an ancestral recipient in Parent 1, forming recombinant ancestor R_{anc}. From R_{anc}, a burst of $K \in \{2, 5, 8, 10, 15\}$ descendant lineages was generated with short terminal branch divergence ($d_{\text{swarm}} = 0.001$), modeling transmission swarms in clinical cohorts (total $N \in \{42, 45, 48, 50, 55\}$, 20 replicates each, 300 alignments, L = 5,000 nt, ground truth $K_{\text{true}} = 1$).
- **Multi-Way Mosaic Swarms**: Backbones comprised 45 background parent taxa evenly divided among three distinct parental clades (15 taxa each for P1, P2, P3) at $d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$. Three multi-way topologies were simulated: (1) *Dual Cassette*: Recombinant lineages acquired a 1,200 nt cassette from P2 (coordinates 1,000–2,200) and a separate 1,200 nt cassette from P3 (coordinates 3,400–4,600) embedded within a P1 background (L = 5,000 nt); (2) *Nested Russian Doll*: An outer 3,000 nt cassette from P2 (coordinates 1,000–4,000) embedded an inner 1,000 nt cassette from P3 (coordinates 2,000–3,000), producing a hierarchical $P_1 \to P_2 \to P_3 \to P_2 \to P_1$ mosaic (L = 6,000 nt); (3) *Triple Crossover Cascade*: Sequential crossovers produced a serial chromosome mosaic composed of P1 (1–2,000 nt), P2 (2,001–4,000 nt), and P3 (4,001–6,000 nt; L = 6,000 nt). Descendant swarms spanned $K \in \{1, 5, 10\}$ (20 replicates each, 540 alignments).
- **Ghost Donor Introgression**: Alignments of length L = 5,000 nt contained 30–40 background parent taxa across three ghost donor regimes ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, $K \in \{2, 5, 10\}$, 20 replicates each, 540 alignments): (1) *Surrogate Sister Clade*: An ancestral lineage diverged into donor D_{ghost} and surrogate sister P_2'; D_{ghost} transferred cassette [1,500, 3,000] to P_1 before being un-sampled, leaving sister P_2' in the alignment; (2) *Pure Ghost Lineage*: The donor lineage branched from the root of the tree and transferred cassette [1,500, 3,000] into P_1 before complete omission, leaving only unrelated contemporary taxa; (3) *Deep Zoonotic Reservoir*: The introgressing tract originated from an outgroup lineage with cross-species evolutionary distance $d_{\text{reservoir}} = 0.15$ into P_1.
- **Bacterial Micro-Conversions**: Alignments (L = 5,000 nt) contained 30 background parent taxa and recombinant swarms ($K \in \{2, 5, 10\}$) across five conversion tract lengths ($L_{\text{tract}} \in \{50, 100, 200, 400, 800\}$ nt, inserted at coordinate 2,100) and three divergence levels ($d_{\text{parent}} \in \{0.02, 0.05, 0.10\}$, 20 replicates each, 900 alignments). The expected number of informative nucleotide substitutions per micro-tract ranged from 1–5 mutations at 50 nt to 16–74 mutations at 800 nt.

Grand-Scale Continuous Parameter Sweep (150,000 Genomes)

To evaluate algorithmic robustness across unconstrained, continuous evolutionary space rather than isolated grid points, we executed a grand continuous parameter sweep of 150,000 whole genomes. Each trial sampled an independent, continuous parameter vector uniformly across evolutionary phase space:

- Alignment Length: $L \sim \text{Uniform}(1{,}000, 30{,}000)$ nt.
- Sample Taxon Count: $N \sim \text{Uniform}(4, 250)$ contemporary genomes.
- Parental Divergence: $d_{\text{parent}} \sim \text{Uniform}(0.005, 0.150)$ substitutions per site.
- Recombination State: Bernoulli trial with $P(\text{Recombinant}) = 0.50$ (75,000 true recombinant trials, 75,000 non-recombinant null controls).
- Recombinant Tract Length: $L_{\text{tract}} \sim \text{Uniform}(50, \min(5{,}000, 0.60L))$ nt.
- Descendant Swarm Size: $K_{\text{swarm}} \sim \text{Uniform}(1, 25)$ descendant isolates.
- Gamma Rate Heterogeneity: Shape parameter $\alpha \sim \text{Uniform}(0.05, 5.00)$.

Simulations were executed across a distributed compute cluster using high-performance compiled Rust binaries with Rayon multi-threading, logging exact per-trial classification accuracy, breakpoint deviation, isolate precision, and wall-clock execution time.

Ground Truth Extraction and Bijective Bipartite Matching

To establish ground truth for reticulation events without relying on heuristic approximations, we computed exact local patristic distance matrices $\mathbf{D}^{(p)} \in \mathbb{R}^{n \times n}$ across every genealogical partition $p \in \{1, \ldots, P\}$ from the branch lengths of tree T_p. For each contemporary taxon i, its local phylogenetic sister at nucleotide $u \in [s_p, e_p)$ is the closest non-identical taxon:

$$\sigma(i, u) = \arg\min_{j \neq i} D_{ij}^{(p)}. \tag{S24}$$

The background Home parent H_i is defined as the majority sister across the full sequence: $H_i = \text{mode}_{u \in [0,L)} \sigma(i, u)$. A true reticulation event occurs when taxon i's sister shifts to a distinct donor lineage $D = \sigma(i, u) \neq H_i$ across a contiguous segment $[u_1, u_2]$. To eliminate trivial non-divergent ancestral coalescences, ground truth events were filtered to require at least one distinguishing mutation between Home and Donor across the tract ($\sum_{u=u_1}^{u_2} \mathbb{I}(mat[H, u] \neq mat[D, u]) \geq 1$).

To score inferred events against ground truth without bias, predicted events $\{\hat{E}_j\}_{j=1}^{\hat{K}}$ were paired to true events $\{E_i\}_{i=1}^{K_{\text{true}}}$ via maximum-weight bijective bipartite matching using the Hungarian algorithm (`scipy.optimize.linear_sum_assignment`). The pairwise affinity matrix $A_{ij} \in [0, 1]$ between true event $i = (C_i, H_i, D_i, u_{1,i}, u_{2,i})$ and inferred event $j = (\hat{C}_j, \hat{H}_j, \hat{D}_j, \hat{u}_{1,j}, \hat{u}_{2,j})$ was formulated as:

$$A_{ij} = S_{\text{triad}}(i, j) \cdot J_{\text{SNP}}(i, j), \tag{S25}$$

where $S_{\text{triad}} = w(C_i, \hat{C}_j) \cdot \frac{1}{2}\left[w(H_i, \hat{H}_j) + w(D_i, \hat{D}_j)\right]$ weights lineage attribution using tree patristic distances with Gaussian kernel width $\sigma = \bar{D}/2$, and $J_{\text{SNP}} = |S_{\text{true}} \cap S_{\text{inferred}}|/|S_{\text{true}} \cup S_{\text{inferred}}|$ is the Jaccard overlap evaluated over distinguishing polymorphic sites between Home and Donor. Unmatched inferred events were penalized as false alarms (reducing precision PPV $= \sum$ matched affinity$/\hat{K}$); unmatched true events reduced sensitivity (TPR $= \sum$ matched affinity$/K_{\text{true}}$).

Empirical Genomic Cohorts and Annotations

Sarbecovirus Whole-Genome Cohort.

We analyzed 83 complete viral genomes (30,956 bp multiple sequence alignment, 13,160 segregating sites) compiled from GISAID and GenBank (Lytras et al., 2022). The cohort spans SARS-CoV-1 (Tor2), SARS-CoV-2 (Wuhan-Hu-1), close bat relatives (Yunnan bat RaTG13, Laotian bat BANAL-52, BANAL-103, BANAL-116, BANAL-236, BANAL-247; Thai bat RacCS203), and pangolin coronaviruses (Guangdong MP789, Guangxi P1E). Gene coordinates follow Wuhan-Hu-1 (NC_045512.2): ORF1ab (nt 266–21,555), RdRp catalytic core (nt 13,468–16,236), Spike glycoprotein (nt 21,563–25,384; NTD codons 1–313, RBD codons 314–540, S1/S2 furin site 23,618 bp, S2 subunit nt 23,619–25,384), ORF3a (nt 25,393–26,220), Envelope (nt 26,245–26,472), Membrane (nt 26,523–27,191), ORF6 (nt 27,202–27,387), ORF7a (nt 27,394–27,759), ORF8 (nt 27,894–28,259), Nucleocapsid (nt 28,274–29,533), and ORF10 (nt 29,558–29,674).

Streptococcus pneumoniae PMEN1 Whole Chromosome.

We analyzed the complete 2.22 Mb chromosome of the pandemic multidrug-resistant Spain23F-1 / PMEN1 lineage of *S. pneumoniae* (ATCC 700669; 11 representative global isolates spanning 1984–2008; Croucher et al. (2011)). Lossless SNP indexing identified 1,516 segregating polymorphic sites across the 2,221,315 bp chromosome. Annotations follow ATCC 700669 (FM211187.1): *pbp2x* (nt 287,466–289,718), capsular polysaccharide locus *cps* (nt 334,103–354,291), *pbp1a* (nt 322,540–324,702), dihydrofolate reductase *folA* (nt 1,623,115–1,623,600), surface antigen adhesin *pspC/cbpA* (nt 2,109,335–2,111,878), and *pbp2b* (nt 1,532,410–1,534,452).

Hominoidea MHC Class I Allelic Cohort.

We analyzed $N = 379$ classical MHC Class I alleles (1,128 bp coding alignment across human, chimpanzee, gorilla, and orangutan; Maccari et al. (2017)). Alleles were retrieved from the IPD-IMGT/HLA and IPD-MHC databases: 150 *HLA-A*, 150 *HLA-B*, and 50 *HLA-C* human alleles; 15 chimpanzee (*Patr*), 8 gorilla (*Gogo*), and 6 orangutan (*Popy*) alleles. Exon annotations follow canonical classical HLA Class I structure: Exon 1 (Leader peptide, nt 1–72), Exon 2 (α_1 domain, nt 73–342), Exon 3 (α_2 domain, nt 343–618), Exon 4 (α_3 domain, nt 619–894), Exon 5 (Transmembrane anchor, nt 895–1,011), and Exons 6–7 (Cytoplasmic tail, nt 1,012–1,128).

Table S1 Quantitative comparison of `RhizAeon` and 3SEQ across the 3,200-trial Posada & Crandall (2001) benchmark suite. Results are grouped by evolutionary regime: False Positive evaluation under Gamma rate heterogeneity ($\alpha \in \{2.0, 0.5, 0.05\}$, $\rho = 0$, 1,200 trials), Clock-like null controls ($\rho = 0$, 400 trials), and Recombinant coalescent trials ($\rho \in \{1, 4, 16, 64\}$, 1,600 trials). Columns report the population mutation parameter (θ), population recombination parameter (ρ), alignment-level power / FPR (%), mean inferred event count ($\hat{K}$), precision (PPV), triad lineage attribution accuracy (S_{triad}), Informative SNP Jaccard spatial overlap (J_{SNP}), and execution runtime per alignment (ms).

		`RhizAeon` (Continuous Manifold Tracking)					3SEQ (Discrete Triplet Scanning)				
Evolutionary Regime	Trials	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)	Power / FPR (%)	$\hat{K}$	Precision	S_{triad}	Time (ms)
A. Rate Heterogeneity Null Suite ($\rho = 0$, Empirical False Positive Calibration)											
$\theta = 10, \alpha \in [0.05, 2.0]$	300	**0.0%**	0.00	100.0%	1.00	19.0	0.0%	0.00	100.0%	1.00	28.2
$\theta = 50, \alpha \in [0.05, 2.0]$	300	**0.3%**	0.00	99.7%	1.00	22.0	0.0%	0.00	100.0%	1.00	26.6
$\theta = 100, \alpha \in [0.05, 2.0]$	300	**0.7%**	0.01	99.3%	0.99	20.9	0.3%	0.00	99.7%	1.00	27.4
$\theta = 200, \alpha \in [0.05, 2.0]$	300	**1.3%**	0.01	98.7%	0.99	24.3	0.3%	0.00	99.7%	1.00	29.2
All Rate-Variation Nulls	*1,200*	**0.58%**	**0.01**	**99.4%**	**0.99**	**21.5**	*0.17%*	*0.00*	*99.8%*	*1.00*	*27.9*
B. Recombinant Coalescent Suite ($\rho > 0$, Statistical Power and Lineage Attribution)											
$\theta = 10$ (Low Divergence)	400	**11.8%**	0.14	17.9%	23.7%	15.1	10.3%	0.26	5.6%	19.7%	31.3
$\rho = 1$	100	3.0%	0.03	55.4%	55.5%	10.9	2.0%	0.04	54.0%	54.2%	41.1
$\rho = 4$	100	13.0%	0.17	24.4%	25.0%	11.9	16.0%	0.40	20.1%	21.2%	37.3
$\rho = 16$	100	26.0%	0.30	9.5%	11.6%	20.2	22.0%	0.58	2.1%	3.2%	23.2
$\rho = 64$	100	5.0%	0.05	2.3%	2.7%	17.5	1.0%	0.03	0.2%	0.3%	23.7
$\theta = 50$ (Moderate Divergence)	400	**67.8%**	1.51	40.7%	54.2%	19.8	83.5%	6.97	16.6%	34.6%	27.6
$\rho = 1$	100	33.0%	0.44	55.2%	58.0%	15.6	38.0%	2.14	41.0%	48.5%	31.7
$\rho = 4$	100	57.0%	0.90	34.3%	37.6%	17.7	73.0%	5.18	7.0%	17.0%	28.9
$\rho = 16$	100	88.0%	2.09	34.3%	45.2%	25.8	96.0%	9.75	6.4%	19.1%	24.5
$\rho = 64$	100	93.0%	2.62	39.1%	58.1%	20.2	100.0%	10.82	11.9%	33.8%	25.4
$\theta = 100$ (Substantial Divergence)	400	**73.5%**	1.87	43.1%	53.2%	21.6	84.3%	10.23	14.2%	29.5%	31.2
$\rho = 1$	100	37.0%	0.48	55.9%	59.7%	15.4	52.0%	3.84	34.9%	46.9%	36.7
$\rho = 4$	100	69.0%	1.20	43.3%	48.4%	18.4	85.0%	8.38	6.3%	17.9%	31.8
$\rho = 16$	100	90.0%	2.29	35.1%	45.1%	22.7	100.0%	14.17	5.4%	19.4%	27.8
$\rho = 64$	100	98.0%	3.51	38.1%	59.5%	29.4	100.0%	14.54	10.3%	33.5%	28.4
$\theta = 200$ (High Divergence)	400	**80.0%**	2.49	45.2%	56.1%	22.5	83.0%	12.12	16.4%	31.3%	36.8
$\rho = 1$	100	43.0%	0.52	64.6%	67.3%	17.1	43.0%	4.05	46.5%	55.2%	39.6
$\rho = 4$	100	83.0%	1.58	38.6%	45.7%	20.4	89.0%	10.92	4.6%	16.7%	37.0
$\rho = 16$	100	96.0%	3.28	39.2%	53.4%	23.4	100.0%	17.25	5.6%	21.2%	37.3
$\rho = 64$	100	98.0%	4.56	38.4%	58.2%	29.4	100.0%	16.26	9.0%	32.6%	33.3

Table S2 Empirical operational phase envelope of `RhizAeon` across 1,000,000 independent continuous simulation trials. Simulations span $L \in [300, 30{,}000]$ nt, $N \in [4, 250]$ taxa, $d_{\text{parent}} \in [0.005, 0.35]$ subs/site, and continuous Gamma rate variation $\alpha \in [0.2, 5.0]$ alongside strict molecular clock controls ($\alpha \geq 50$). Multi-event topologies include true nulls ($K = 0$), 5'/3' terminal crossovers, intercalary conversion cassettes ($K = 1$), and multi-donor / multi-event mosaicism ($K \in \{2, 3, 4\}$). Breakpoint errors report median absolute error (nt) among detected recombinant tracts.

Evolutionary Regime	Parameter Space	Total Trials	Detected / Rec	Power / FPR	Median Jaccard	Median BP Error
Strict Molecular Clock Null	$d \in [0.005, 0.35], \alpha \geq 50$	83,311	271	0.325% FPR	1.000	—
Gamma Heterotachy Null	$\alpha \in [0.2, 5.0], d \in [0.005, 0.35]$	250,023	919	0.368% FPR	1.000	—
Taxon-Level False Alarm	$N_{\text{taxa}} = 22,744,905$ total leaves	333,334	2,651 taxa	0.0117% FPR	—	—
Recombinant Partition Stratified by Informative Tract Mutations (M_{tract})						
Tract SNPs $M \in [0 - -2]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	85,564	402	0.5%	0.001	2,150.0 nt
Tract SNPs $M \in [3 - -5]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	68,609	4,678	6.8%	0.041	20.0 nt
Tract SNPs $M \in [6 - -10]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	77,568	22,646	29.2%	0.203	23.0 nt
Tract SNPs $M \in [11 - -15]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	52,607	27,587	52.4%	0.387	25.0 nt
Tract SNPs $M \in [16 - -25]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	71,427	51,096	71.5%	0.553	23.0 nt
Tract SNPs $M \in [26 - -50]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	96,694	82,714	85.5%	0.705	17.0 nt
Tract SNPs $M \in [51 - -100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	84,676	77,213	91.2%	0.775	13.0 nt
Tract SNPs $M \in [> 100]$	$L_{\text{tract}} \in [45, 0.45L]$ nt	129,521	123,692	95.5%	0.766	9.0 nt
Multi-Event and Structural Topology Decomposition ($K_{\text{true}} \geq 1$, Total $N = 666,666$)						
Single Event ($K_{\text{true}} = 1$)	Crossover / Cassette	366,396	208,985	57.0%	0.418	15.0 nt
Dual Events ($K_{\text{true}} = 2$)	Two Reticulations	166,279	101,350	61.0%	0.522	14.0 nt
Triple Events ($K_{\text{true}} = 3$)	Three Reticulations	80,391	48,288	60.1%	0.502	15.0 nt
Quad Events ($K_{\text{true}} = 4$)	Complex Mosaicism	53,600	31,405	58.6%	0.476	16.0 nt
Terminal Crossovers	5' or 3' Exchange	239,767	171,194	71.4%	0.548	12.0 nt
Internal Cassettes Only	Intercalary Swap	426,899	218,834	51.3%	0.422	18.0 nt